

SHTPC: Stateful Hidden Trajectory-Payload Consistency for Private Heatmap Aggregation

Yingjie Wang, Hao Wang, Wenhan Hou, Haojun Teng, Kaile Su, Zhipeng Cai

Abstract—Managed heatmap services need to screen location reports before private aggregation without exposing coordinates or primary cells. Existing compositions can validate a hidden trajectory and an encoded payload separately, yet no single relation guarantees that both refer to the same primary cell. We introduce Stateful Hidden Trajectory-Payload Consistency (SHTPC), a stateful Groth16 admission relation. It binds accepted-predecessor continuity, window reachability, the coordinate-derived primary, and A/R payload commitments within one report context. The evaluated circuit has 3,547 constraints, 16 public inputs, and 25 private inputs, and it passes all 22 static checks. Across 15 $N=1000$ dataset-seed units, 225,000 proofs verify with 0 failures. Under one fixed three-seed configuration, SHTPC yields positive released-set Jaccard differences against a mechanism-level Nebula-style adapter on all five datasets, although two differences are small. SHTPC proves post-enrollment proof-to-payload consistency. It does not establish physical presence, eliminate metadata leakage, or reject false reports that remain inside the public mobility envelope.

Index Terms—Mobile crowdsensing, Private aggregation, Zero-knowledge proofs, Admission consistency, Secure aggregation.

I. INTRODUCTION

MANAGED private heatmap services turn enrolled-device reports into population-level spatial statistics for traffic engineering, public transit planning, and environmental monitoring [1, 2]. Such services can be viewed as a managed form of mobile crowdsensing, where enrollment and admission state are available but raw coordinates and primary heatmap cells should remain hidden. Such deployments often require three properties at once. A verifier should admit only reports that are consistent with a public mobility envelope. The verifier should not learn exact coordinates or primary heatmap cells. The aggregation roles should receive only encrypted or share-encoded payloads before the final differentially private release [3–8]. This setting is narrower than fully anonymous trajectory

publication. It assumes enrollment, persistent pseudonymous admission state, and separated aggregation roles.

The structural gap is a proof-to-payload gap. A coordinate-hiding system may validate a hidden trajectory proof and a heatmap payload through different artifacts. The proof can show that a hidden coordinate x_t follows the previous accepted hidden state and remains inside a public K -window envelope. The payload can separately pass encoding, context, or share-validation checks. If these checks use independent private primary-cell witnesses, the system may accept a proof generated for one hidden primary while forwarding a payload that contributes to another primary cell. The Shuffler cannot see the coordinate, and an Aggregator cannot see the proof witness. Neither role can detect the mismatch unless the coordinate-derived primary and the routed payload are bound within the same accepted relation.

A concrete adversarial report illustrates the problem. The client proves admissibility for a hidden coordinate whose primary cell is p_t . It then substitutes encrypted or share-encoded heatmap material whose one-hot contribution is assigned to $p'_t \neq p_t$. Context hashes and transport authentication can bind a report frame, but they do not establish that the payload primary was derived from the same hidden coordinate used in the trajectory proof. Preventing this attack under coordinate hiding requires one of three choices. The system can reveal the primary to a checking role, rely on a trusted plaintext checker, or introduce an in-relation binding between the coordinate-derived primary and the encoded payload. This paper studies the third choice.

SHTPC addresses this gap as a stateful zero-knowledge admission layer placed before private heatmap aggregation. Each submission carries a compact SHTPCMAIN Groth16 proof that links the current hidden location to the previous accepted state, enforces per-step speed bounds and Anchor Distance Window Continuity (ADWC), derives the hidden primary heatmap cell, and binds that primary to report and A/R share-content commitments under the current report context. The Shuffler verifies proof validity, chain continuity, timing tokens, and transport authentication before forwarding accepted reports to secure aggregation. The downstream ϵ -DP heatmap release is unchanged [9].

The claim is intentionally limited to post-enrollment data-plane consistency. SHTPC does not prove physical presence, prevent false genesis at enrollment, prevent voluntary credential or device transfer, provide trajectory anonymity, secure a malicious Shuffler, or reject reports that remain fully inside the public envelope. The Shuffler also learns L_S , including chain topology, timing, mode or tier metadata, and accept or

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reject outcomes. These signals are not covered by the public heatmap’s $\varepsilon_{\text{total}}$ -DP accounting. The security contribution is therefore the SHTPC admission-consistency property itself. Groth16, commitments, ADWC, share digests, and DP release are building blocks for realizing that property under explicit leakage and trust boundaries.

This paper makes three contributions.

- Formalizes SHTPC as a stateful post-enrollment admission-consistency property for coordinate-hiding heatmap aggregation, with explicit predecessor, K -window, hidden-primary, payload-binding, leakage, and corruption boundaries.
- Designs a zero-knowledge admission layer that binds these predicates in one accepted statement. The paper further proves separation lemmas showing why stateless ZK, VDAF-style validation, context commitments, and post-admission defenses do not imply SHTPC, and gives an end-to-end payload-consistency theorem connecting the circuit witness, public binders, ciphertext blobs, Aggregator validation, and the final clipped one-hot contribution.
- Evaluates SHTPC as admission-boundary enforcement, with design-space baselines, measured proof-to-payload substitutions, state/window-boundary checks, metadata leakage analysis, utility cost, and reproducible artifact evidence under the stated deployment boundary. The fixed comparison has a positive observed Jaccard difference for every one of the 15 dataset-seed pairs, although two dataset means differ only slightly. The full-path matrix verifies 225,000 proofs with 0 failures.

II. RELATED WORK

SHTPC is closest to work that protects location analytics, validates private aggregate inputs, and proves predicates over hidden data, but those systems target different consistency boundaries. Dwork et al. protect released statistics with differential privacy [3]. Bittau et al. separate encoding, shuffling, and analysis in Prochlo [4]. Corrigan-Gibbs and Boneh validate secret-shared measurements in Prio [7]. Tippenhauer et al. study how a location claim can be spoofed or authenticated [10]. Our work instead defines a post-enrollment admission relation for private heatmap aggregation. The verifier accepts a hidden report only when its predecessor state, K -window trajectory envelope, coordinate-derived primary cell, and routed aggregation payload are mutually consistent. Table I summarizes this boundary.

A. Comparison with Existing Families

This comparison fixes the scope of the contribution. SHTPC is not a new VDAF, differential-privacy mechanism, robust statistic, proof-of-location protocol, or proof system. It is an admission layer that rejects hidden reports whose accepted state, reachable envelope, derived primary, and routed payload are inconsistent. The STATEFUL-CONTEXTCOMMIT row is the measured composition harness used in Section VI-C. It captures a natural state-and-context composition that still keeps trajectory and payload validity as separated statements.

TABLE I
COMPARISON WITH ADJACENT SYSTEM FAMILIES.

Family	Boundary relative to SHTPC
Prio3/VDAF [7, 8]	Validates encoded measurements, but not accepted-predecessor continuity, K -anchor reachability, or same-primary trajectory-payload equality.
Prochlo/Nebula [4, 11]	Separates encode, shuffle, and analyze roles, but admission remains local to a submitted report and visible metadata remains a system boundary.
Private location-service accounting [12–14]	Audits or prices location-dependent records, but does not provide payload-bound private heatmap admission.
Trusted plaintext filter	Can check coordinate and payload equality if it sees coordinates, but does not preserve coordinate hiding at admission.
Stateless ZK filter [15, 16]	Can prove one hidden predicate, but lacks accepted-state continuity and K -window drift control.
Robust aggregation [17–20]	Mitigates aggregate poisoning after inputs are admitted, but does not prove hidden report consistency before secure aggregation.
Proof-of-location [10]	Establishes presence or proximity, not payload-bound private heatmap admission.
STATEFUL-CONTEXTCOMMIT	Tracks state and context, but no same-primary equality is proved in one relation.
SHTPC	Accepted predecessor, K -anchor, hidden-primary derivation, and payload binding under explicit L_S .

B. Private Location Analytics

Private location analytics primarily protects either the published statistic or the submitted location representation. Differential privacy, local/shuffle privacy, geo-indistinguishability, trajectory anonymization, and private trajectory synthesis reduce what is revealed by outputs or traces [3, 21–43]. These mechanisms are compatible with a heatmap release and can be used downstream of SHTPC. Their trust boundary is nevertheless output privacy or trace obfuscation, not admission consistency. They do not decide whether a hidden coordinate is reachable from the previously accepted hidden state or whether its primary cell is the one routed into aggregation.

C. Private Aggregation and Input Validation

Private aggregation and validation systems protect messages, shares, or encoded measurements after collection. Prochlo, Nebula, secure aggregation, Prio, Poplar, ACORN, and VDAF-style protocols separate collection roles and validate encoded measurements [4–9, 11, 44–50]. Robust aggregation and federated-learning integrity defenses reduce the effect of malicious inputs after admission [15, 17–20, 51–55]. These systems are essential once an input has been admitted, but their validation predicates are normally measurement-local or aggregate-robustness oriented. SHTPC addresses the preceding admission boundary. The hidden trajectory relation and heatmap payload must be one accepted statement before secure aggregation separates the report across roles.

D. Location Authenticity and Private Accounting

Location authenticity and private accounting systems provide useful deployment building blocks but target different predicates. GPS-spoofing defenses, GNSS authentication, anonymous credentials, trusted processors, and secure elements can strengthen enrollment or possession claims [10, 56–62]. VPriv, PrETP, PriPAYD, and private tolling protocols combine hidden location evidence with charging or audit rules [12–14, 63–66]. These lines of work can improve how a client

is enrolled, how a device anchors its sensor evidence, or how a private service audits billing outcomes. Their predicates do not express repeated heatmap admission with predecessor continuity, K -window reachability, and same-primary payload binding.

E. Zero-Knowledge Proofs and Circuit Binding

Zero-knowledge proof systems provide the substrate for checking private relations without revealing witnesses [16, 67–80]. For SHTPC, the central question is not merely proof size or verifier time, but which equalities are enforced inside one relation. Prior analyses show that missing constraints can create practical ZK soundness bugs [81]. Accordingly, SHTPC’s C17–C21 constraints derive the hidden primary from the hidden coordinate and bind that primary to the report-level payload and A/R share commitments under one report context.

III. SHTPC PROBLEM DEFINITION AND COMPOSITION GAPS

This section defines the admission problem addressed by SHTPC. The goal is not to prove physical presence or trajectory anonymity. The goal is to decide whether a hidden report may enter private heatmap aggregation while preserving coordinate and primary-cell hiding under the stated Shuffler leakage.

A. Report and Admission Model

The notation uses uid for the enrolled pseudonymous chain handle, e for the enrollment or reset epoch, t for the admission round, and w for the heatmap release window. The hidden coordinate at round t is $x_t = (x, y)$, its derived heatmap cell is $primary_t$, and L_S denotes Shuffler-visible leakage.

The monitored region is partitioned into a public $W \times H$ grid with cell size s . A hidden coordinate determines

$$primary_t = W \lfloor y_t/s \rfloor + \lfloor x_t/s \rfloor.$$

where W is the grid width in cells and $D = WH$ is the number of heatmap cells. The evaluated grid uses $s = 100$ m, $W = 100$, and $H = 100$. In each release window, an accepted client contributes a clipped one-hot vector $\tilde{\mathbf{u}}_w^{(i)} \in \mathbb{R}^D$ with $\|\tilde{\mathbf{u}}_w^{(i)}\|_1 \leq 1$. The aggregate target is $\mathbf{s}_w = \sum_i \tilde{\mathbf{u}}_w^{(i)}$. SHTPC does not modify this target or the downstream pure $\varepsilon_{\text{total}}$ -DP heatmap release [3]. It only decides whether a hidden report is admissible before aggregation. This DP guarantee applies to the released heatmap only. It does not protect L_S metadata.

SHTPC uses salted commitments and a zero-knowledge admission relation to check hidden location consistency without exposing coordinates or primary cells. The concrete Groth16, Poseidon, and MiMC7 instantiations are specified in Sections IV-C and IV-H, and the corresponding assumptions are used in Section V [16, 78, 79].

B. Stateful Hidden Trajectory-Payload Consistency

The central property is stateful because the Shuffler stores the last accepted hidden state for each enrolled identity. It is hidden because the checked coordinate and primary cell remain circuit witnesses.

Definition 1 (Anchor Distance Window Continuity (ADWC) Predicate). For trajectory $\{p_i\}_{i \geq 0}$ with $p_i = (x_i, y_i)$, window parameter $K \geq 1$, anchor $a_i = p_{\max(0, i-K)}$, tier speed τ_u , Shuffler-issued interval Δt_i , window time budget W_i , and deployment cap $\text{cap}_{\text{policy}}$, ADWC holds at step i iff

$$\begin{aligned} \|p_i - p_{i-1}\|^2 &\leq \tau_u^2 \Delta t_i^2, \\ \|p_i - a_i\|^2 &\leq \min(\tau_u^2 W_i^2, \text{cap}_{\text{policy}}^2). \end{aligned} \quad (1)$$

where $W_i = K \Delta t_{\text{slot}}$ in the fixed-slot artifact evaluated in this paper, and a variable-slot deployment must instead use a Shuffler-signed window-level time sum. For $i < K$, $a_i = p_0$, so warmup uses the same circuit path rather than skipping anchor checks.

Definition 2 (Stateful Hidden Trajectory-Payload Consistency (SHTPC)). An accepted post-enrollment report satisfies SHTPC if five conditions hold for the same hidden report witness. First, the predecessor commitment opens to the previous accepted Shuffler state for the same uid and epoch. Second, the current hidden coordinate satisfies ADWC with respect to the predecessor and the K -window anchor. Third, the primary heatmap cell is derived from the hidden coordinate by the public grid rule. Fourth, the submitted aggregation payload is bound to that same hidden primary cell under the report context. Fifth, uid , epoch, round, timing token, and report context match the accepted public statement.

The SHTPCMAIN relation realizes SHTPC. It proves knowledge of hidden coordinates, commitment salts, the primary cell, payload-binding openings, and the enrollment secret such that

$$\begin{aligned} \mathcal{R}_{\text{SHTPC}} \quad &\ell_t = \text{Poseidon}(x_t, y_t, r_t), \\ &\ell_{t-1}, \ell_a \text{ open accepted predecessor} \\ &\text{and anchor states,} \\ &\|p_t - p_{t-1}\|^2 \leq \tau_u^2 \Delta t_t^2, \\ &\text{ADWC}(p_t, p_a, K) = 1, \\ &primary_t = W \lfloor y_t/s \rfloor + \lfloor x_t/s \rfloor, \\ &(\text{Com}_{\text{pri}}, \text{Com}_{\text{pay}}, \text{Com}_A, \text{Com}_R) \\ &\text{bind the same } primary_t, \\ &\text{Poseidon}(\text{secret}, uid) = \text{com}_{\text{sec}}, \\ &\text{SRV-TS}(uid, \text{epoch}, t) = 1, \\ &\text{ctx}_t = (uid, \text{epoch}, t, \text{token}, \text{task}), \\ &\text{ctx}_t \text{ matches all public binders and blobs.} \end{aligned} \quad (2)$$

The Shuffler checks the public predecessor state, timing token, and transport hash around this relation. The Aggregators validate only their channel-local share packages.

C. Leakage Boundary and Goal Scope

Definition 3 (Explicit Shuffler Leakage Function). The Shuffler-visible leakage L_S contains the uid or pseudonymous

TABLE II
CLAIM AND CORRUPTION BOUNDARY FOR SHTPC.

Claim or profile	Boundary
Raw coordinate hiding	Game 3, equal- L_S single-server views.
Primary hiding	Game 3, equal- L_S . Metadata remains visible.
HBC Shuffler	Sees L_S , including metadata that is not DP-protected.
One Aggregator	Sees one share package. Single-share privacy applies.
Trajectory anonymity	Not claimed.
Report unlinkability	Not claimed because per-uid state is required.
Output heatmap	Pure $\varepsilon_{\text{total}}$ -DP only for released statistics.
A+R collusion	Outside input-privacy claim. Output DP remains.
Malicious Shuffler	Outside claim. Requires audit or replicated logs.
False genesis/transfer	Deployment scope.
In-envelope false reports	Residual risk accepted by design.

chain handle, epoch, round, timing slot, interarrival pattern, chain edge, mode or tier tag, proof outcome, accept or reject outcome, reset or warmup status, message-size class, and public context identifiers. It excludes raw coordinates and primary cells, but it includes enough topology and timing information to reveal behavioral patterns. SHTPC therefore proves coordinate-and-primary hiding only for equal- L_S worlds. It does not provide trajectory anonymity or report unlinkability.

SHTPC uses four goal labels. In Game 1, an adversary wins if a report is accepted while its extracted witness violates predecessor consistency, per-step continuity, or the ADWC anchor predicate. In Game 2, an adversary wins if a report is accepted and the forwarded share package is bound to a primary cell different from the primary cell derived from the hidden coordinate in the accepted trajectory proof. In Game 3, two executions with equal L_S , chain topology, timing bins, message-size classes, and accept/reject traces must be indistinguishable with respect to hidden coordinates and primary cells for an allowed single-server view. In Game 4, the released heatmap satisfies output DP only. After per-user, per-round L_1 clipping, the sequence of released heatmaps satisfies pure $\varepsilon_{\text{total}}$ -DP under user-level adjacency. This goal applies only to the public heatmap release, not to L_S or other protocol-control metadata. Section V gives the formal bounds.

Admission functionality. SHTPC is modeled as an admission functionality $\mathcal{F}_{\text{SHTPC-Adm}}$ before private aggregation. For each (uid, e) , it stores accepted state c_t , a round counter, an anchor pointer, and warmup status. A submission is accepted and forwarded only if Definition 2, SRV-TS, SCS, and transport authentication all hold. Warmup updates state but does not contribute payloads. Reset creates a new epoch. The Shuffler receives exactly L_S . A single Aggregator receives only its channel-local share package and validation outcome. The Decoder receives aggregates for downstream DP release. This is a standalone admission model, not UC security for the complete deployment.

Enrollment consistency and Sybil cost are deployment propositions rather than core cryptographic games. A valid enrollment tuple $(uid, \text{tier_vmax_sq}, \text{com}_{\text{sec}}, \text{com}_{\text{modeset}})$ must be bound to threshold-valid EA attestations. The number of simultaneous Sybil identities is bounded only by the adversary’s external credential budget and the deployment’s enrollment

controls.

D. Violation Families and Coverage

The compact matrix below summarizes the concrete SHTPC-violation families used throughout the paper and separates circuit soundness from deployment-scope controls.

Family	Broken invariant	Coverage
A1 teleportation	Step continuity	Theorem 1
A2a substitution	Enrolled secret	Lemma 6
A2b transfer	Possession	Discussion
A3 gradual drift	K -window ADWC	Theorem 1
A4 Sybil creation	Enrollment	Proposition 1
A5 replay or reuse	Shuffler state	Theorem 1
A6 payload mismatch	Same primary	Theorem 2

E. Why Natural Compositions Are Insufficient

This subsection states why SHTPC is not obtained by placing an ordinary local validity proof before a private aggregation protocol. The statements are predicate-omission separations rather than impossibility theorems. If an admission relation omits a predicate required by SHTPC, then there is an accepted adversarial construction that violates the corresponding SHTPC condition while satisfying the predicates that remain.

Lemma 1 (Stateless local validity does not imply predecessor consistency). *Consider any admission protocol whose verification predicate at round t is independent of the previously accepted commitment state for the same uid and epoch. Then there exists a malicious client that produces individually valid reports whose accepted hidden states do not form a valid predecessor chain. Therefore stateless local validity does not imply SHTPC condition (i).*

Sketch. Let x and x' both satisfy the local range, grid, and payload-format predicates, while their commitments are not linked by the current Shuffler state. If the predicate omits C_{t-1} , the report for x' is accepted whenever the local predicate holds. The accepted sequence can violate predecessor consistency while every local proof verifies.

Lemma 2 (Step-only continuity does not imply K -window envelope consistency). *Per-step continuity does not imply the K -window ADWC predicate. For any protocol that checks only $\|x_i - x_{i-1}\| \leq \tau\Delta t$, there exists a sequence of locally valid steps whose displacement from the K -window anchor exceeds a public deployment cap.*

Sketch. Let $x_i = x_0 + i \cdot \tau\Delta t \cdot e$ for a unit direction e . Each transition satisfies the per-step bound. After K steps, $\|x_K - x_0\| = K\tau\Delta t$, which can exceed a public cap even though every adjacent transition is locally valid.

Lemma 3 (Separated trajectory and payload statements do not imply same-primary binding). *Independent trajectory and payload-validity statements do not imply same-primary payload binding. If the trajectory relation and payload relation use independently chosen primary witnesses, or expose only separate commitments without an in-circuit equality constraint, then there exists a proof-payload mismatch adversary.*

Sketch. The adversary proves trajectory admissibility for a hidden coordinate x whose derived primary is $p = \text{Grid}(x)$. It then constructs an aggregation payload committed or routed to $p' \neq p$. Without a common committed primary and in-circuit equality, both statements can verify.

Lemma 4 (Context-only validation does not express SHTPC). *Context-bound measurement validation does not imply SHTPC unless the context-validity predicate includes the accepted predecessor, K -window anchor, hidden coordinate-to-primary derivation, and same-primary equality relation.*

Sketch. Context validation can bind an encoded measurement to public metadata such as epoch, task identifier, aggregation parameters, or blob hashes. Without a hidden state-transition predicate and a common primary witness shared by trajectory and payload relations, it validates report format or context, not SHTPC.

These separations motivate a single accepted relation that binds predecessor state, the K -window anchor, the hidden coordinate, the coordinate-derived primary, and the routed payload. They also guide the evaluation design. We measure the SHTPC row, mechanism ablations, and the strongest low-cost stateful context-commitment harness. TEE and VDAF variants are treated as analytical baselines because they either change the privacy boundary or become equivalent to adding an SHTPC-style in-relation primary binding.

IV. METHOD

SHTPC follows the Encode-Shuffle-Analyze (ESA) pipeline. It inserts a proof-carrying admission layer at the Shuffler before forwarding to secure aggregation. The protocol involves five role types, namely Client, Enrollment Attestation Authority (EA), Shuffler, two non-colluding Aggregators A and R, and Decoder. The design objective is to realize SHTPC (Definition 2), not to replace the downstream private aggregation or DP release. Section III-E explains why natural stateless, context-only, or separated-proof compositions miss at least one SHTPC condition.

A. System Overview

Figure 1 shows the full pipeline. Each report first passes Shuffler-side proof and state verification. Accepted reports then contribute additive shares to Aggregators A and R, and the Decoder releases their aggregate as a DP heatmap. The rest of this section details the commitment chain and payload binders (§IV-C), the distance-verification circuit (§IV-D), enrollment and warmup (§IV-E), and the differential privacy layer (§IV-F).

In each reporting round, a client encodes its observation into additive shares of a clipped one-hot heatmap contribution. The primary cell is derived from the hidden coordinate. The client updates its commitment chain and generates a compact SHTPCMAIN proof. The proof binds the current hidden location, predecessor location, anchor location, identity commitment, tier/mode signals, hidden primary commitment, report-level payload commitment, and A/R share-content commitments while proving ADWC with zero-knowledge privacy.

Additive secret sharing [82] between the two Aggregators ensures that neither observes plaintext contributions alone. The implementation uses seed-compressed fixed-layout shares with VDAF-style expansion checks for single-Aggregator index hiding. Commitments are hash-based salted commitments instantiated with Poseidon over BN254 for efficient zero-knowledge proof circuits.

The Shuffler maintains an append-only per-uid, per-epoch accepted-state log under SCS. Accepted reports update the chain only after proof verification, timing-token validation, predecessor matching, payload-binder validation, and transport-authentication checks pass. Enrollment and reset enter warmup, where the same circuit path advances state but accepted payloads are withheld from aggregation.

B. Threat Model

The threat model considers malicious clients that may deviate arbitrarily from the protocol and may coordinate with one another. They know all public parameters, including tier speed bounds, ADWC caps, the window parameter K , and the verification key. They do not know honest clients' device-held secrets or raw coordinates.

The adversary may attempt the SHTPC-violation families A1 to A6 defined in §III-D. The circuit and Shuffler state address A1, A2a, A3, A5, and A6. A4 relies on EA threshold attestation against Sybil collusion [83]. A2b and false enrollment genesis points are deployment-scope issues handled by external controls rather than by the SNARK relation.

The server side follows the separated-role design of Prochlo and Nebula [4, 11]. The Shuffler is honest-but-curious. It observes proofs, commitments, ciphertext metadata, and accept/reject outcomes, yet it never sees plaintext coordinates. Aggregators A and R are further assumed not to collude. This organizational separation is an explicit deployment assumption. It is not implied by regulatory compliance or by assigning different service names to the roles. These are claim boundaries rather than engineering conveniences. If the Shuffler forks or rolls back state, the envelope-admission theorem is not claimed without an external audit log or replicated state machine. If the Shuffler colludes with an Aggregator, Game 3 no longer gives the stated single-server input-hiding view. If both Aggregators collude, input privacy degrades to the public output-DP guarantee. If the timing service lets clients choose Δt , the motion-envelope statement is not claimed.

Timing follows a server-issued assumption called SRV-TS. Under SRV-TS, the Shuffler honestly issues and verifies fresh time-slot tokens for each $(uid, epoch, w)$, so clients are unable to choose, replay, or inflate $step_dt_sq$ and prove only against a Shuffler-issued value. The token binds the user, epoch, round, previous accepted server time, current coarse server slot, squared interval, and nonce. This binding avoids relying on an unpredictable packet receive timestamp during proof generation while still preventing clients from selecting their own Δt .

Assumption SCS. Shuffler State Consistency. For each $(uid, epoch)$, the Shuffler maintains an append-only accepted-state log with a monotonic round counter. A report is accepted

TABLE III
DESIGN-SPACE BASELINES FOR THE SHTPC ADMISSION BOUNDARY. THE TABLE SEPARATES INCOMPLETE BUT PLAUSIBLE COMPOSITIONS FROM ALTERNATIVES THAT CHANGE THE PRIVACY OR TRUST MODEL.

Design	Coord. hidden	Stateful	Payload bound	Primary substitution	Cost or limitation
Stateless ZK + same-primary check	Yes	No	Partial	Partial	It can bind one report's proof and payload, but it does not express accepted predecessor state, replay prevention, or K -anchor consistency.
Stateful context commitment	Yes	Yes	Partial	No	The frame and blob hash are bound, but trajectory and payload statements may use different private primary witnesses.
TEE or plaintext filter	No or partial	Yes	Yes	Yes	It checks the predicate by trusting a runtime or revealing coordinates or primary cells to a checker. This is a privacy-cost upper bound.
VDAF/Prio with visible primary	No primary hiding	Partial	Yes	Yes	Validation can bind the encoded primary, but the primary is visible to the validation path.
VDAF/Prio with hidden unbound primary	Yes	Partial	No	No	Hidden measurement validation does not force the payload primary to equal the trajectory-derived primary.
VDAF/Prio with relation-bound primary	Yes	Depends	Yes	Yes	Preventing substitution requires an equivalent in-relation primary binding, which is the SHTPC subrelation studied here.
SHTPC	Yes	Yes	Yes	Yes	It binds accepted predecessor state, K -anchor reachability, coordinate-derived primary, and routed payload under explicit L_S .

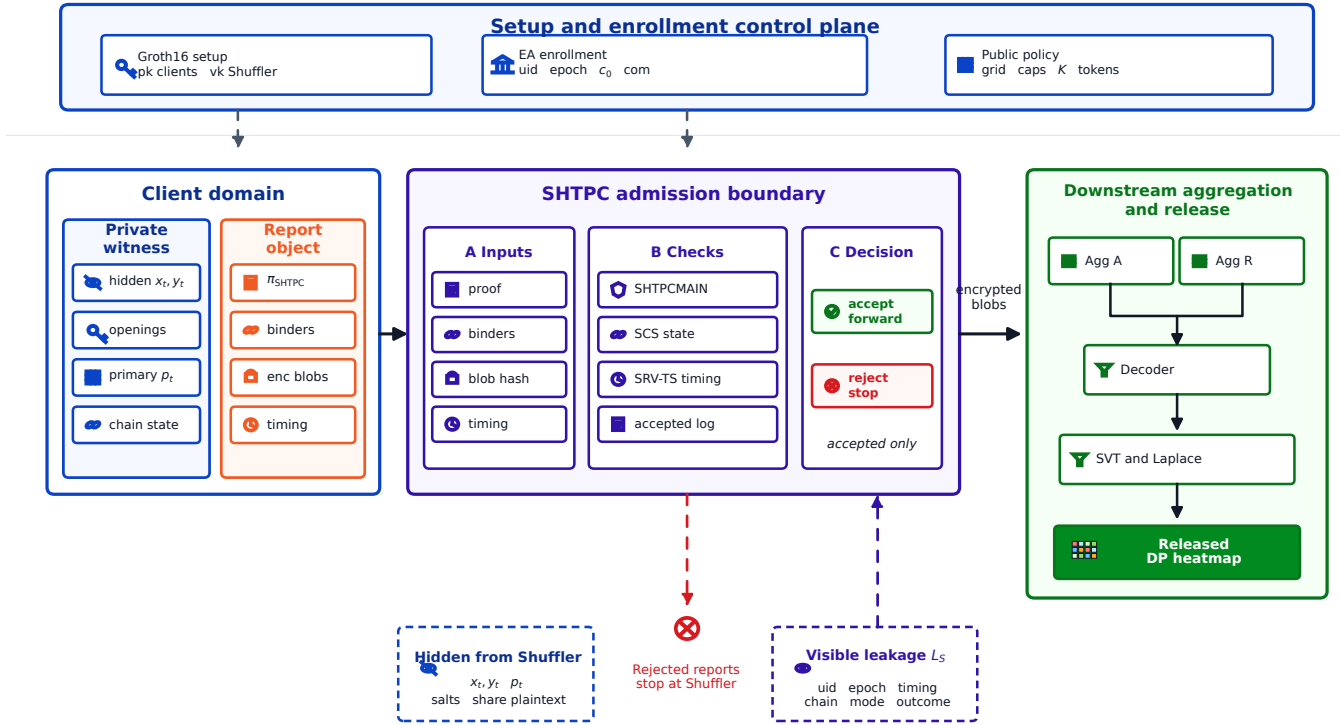


Fig. 1. Overview of SHTPC. Setup, enrollment, and public policy define the control plane. Each client report carries a proof, public binders, and encrypted payload blobs. The Shuffler enforces the admission boundary and forwards only accepted encrypted blobs to secure aggregation and downstream DP release. The lower callouts separate hidden witness values from Shuffler-visible leakage L_S .

only through an atomic compare-and-swap transition from the current predecessor state to a unique successor state, and the accepted record is durably logged before the payload is forwarded. Crash recovery restores the maximum durable accepted round. Duplicate rounds, forked successors, stale predecessors, and rollback attempts are rejected. Reset creates a new epoch and re-enters warmup. The admission soundness theorem relies on SCS and does not cover a malicious Shuffler that violates this state discipline.

Enrollment authorization follows assumption CA3. Under

CA3, at least t out of n enrollment anchors are non-colluding. The default profile is $t = 3$, $n = 5$, and a single-operator deployment is the degenerate case $t = 1$, $n = 1$.

SHTPC deliberately leaves two issues to the deployment layer. They are ground-truth enrollment (c_0) and voluntary credential or device transfer (A2b). These risks fall outside Theorems 1–2 and require proof-of-location at enrollment, device attestation or secure-element sealing, re-enrollment monitoring, and behavioral risk scoring.

Even under credential or device transfer, the transferred identity still must satisfy the public trajectory envelope, pay-

load binding, timing tokens, and chain continuity. Proposition 1 gives a deployment cost bound for creating additional accepted identities, but it does not prevent a legitimate user from voluntarily transferring an existing credential or device.

Warmup raises the cost of false-genesis by requiring $N_{\text{warm}} = \max(K, 2)$ plausible rounds before aggregation. Collusion of both Aggregators lies outside Game 3's single-Aggregator privacy guarantee, in which case input privacy degrades to the public output DP layer while Shuffler-side admission checks still run.

Independent operation can make the non-collusion assumption auditable, but it cannot enforce that assumption cryptographically. A conforming deployment must therefore place the Shuffler and Aggregators in distinct administrative domains, document their data access, and audit state transitions and cross-role flows. If one operator controls or can silently correlate these roles, the single-server privacy claim does not apply.

The component leakage boundary is summarized in Table II. The Shuffler sees SHTPCMAIN, public payload binders, blob_hash, fixed-size ciphertext metadata, timing tokens, and uid/epoch metadata, but not share plaintext, primary cells, or raw coordinates. Each Aggregator sees only its own decrypted share blob, round context, and validation outcome. The implementation invariants are fixed ciphertext length, fixed-layout expanded shares after local decoding, no primary-index logs, and Shuffler routing without share decryption.

C. Commitment Chain and Payload Binders

The commitment chain and payload binders provide the state-and-payload substrate needed by SHTPC. The chain makes the predecessor relation server-checkable without revealing coordinates, while the binders give the circuit a common handle for connecting the hidden primary cell to the routed share material.

Each client maintains a chain $\{c_w\}_{w=0}^T$. The per-round location commitment is $\ell_w = \text{Poseidon}(x_w, y_w, r_w)$. Its salt is a circuit witness and is never exposed as a public input.

For compact notation, this paper writes Com_{pri} for the hidden primary commitment, Com_{pay} for the report-level payload commitment, and $\text{Com}_A, \text{Com}_R$ for the routed A/R share-content commitments. These binders are public inputs to SHTPCMAIN. Their openings are private witnesses.

The chain commitment is

$$c_w = \text{MiMC7}(c_{w-1}, \ell_w, t_w, w, \text{Com}_{\text{pri}}, \text{Com}_{\text{pay}}, \text{Com}_A, \text{Com}_R, \text{com}_{\text{sec}}), \quad (3)$$

where c_{w-1} is the previous chain value, t_w is the Shuffler-issued timestamp slot, w is the round index, and the remaining terms are the public payload and identity binders for the same report.

The circuit derives the primary cell from the hidden coordinate exactly as in §III ($\text{cell}_x = \lfloor x_w/s \rfloor$, $\text{cell}_y = \lfloor y_w/s \rfloor$, and $\text{primary} = \text{cell}_y \cdot W + \text{cell}_x$ with $s = 100\text{m}$, $W = 100$, and $H = 100$). The primary cell and share-content digests are circuit witnesses, not public inputs. The circuit commits

the hidden primary and binds the same committed primary to the report-level payload commitment and to the A/R share-content commitments under the current report context. The full binding relation (C21) is specified in Section IV-H. The key property is that all payload binders are derived from the same hidden primary and report context inside a single SHTPCMAIN statement.

The client sends seed-compressed fixed-layout share material with VDAF-style checks to Aggregator A and Aggregator R through separate ciphertext blobs. It publishes $\text{blob_hash} = H(\text{blob}_A, \text{blob}_R, \text{ctx})$ for transport authentication. The hash H is collision resistant, and ctx is the report context. The Shuffler recomputes and verifies this digest against the proof public input to prevent post-proof ciphertext-package substitution. On receipt, each Aggregator decrypts its own blob and validates the encoded share package. The checks cover length, domain separator, context fields, and deterministic expansion to a D -cell layout. Aggregator-side validation is a format and context check. The hidden-primary semantic binding is enforced inside the circuit by SHTPCMAIN (C21). **Share representation and frame size.** The privacy claim refers to the expanded aggregation view rather than to an uncompressed wire vector. After local expansion, every cell position has a share, and any one Aggregator's per-cell shares are indistinguishable from random without the other Aggregator's share. On the wire, the prototype sends seed-compressed encrypted share envelopes plus public proof material rather than a D -element wire vector. The prototype does not report an aggregate frame size because no dedicated serialization benchmark covers the complete report object.

Report frame and payload validation syntax. Each submitted report is represented as

$$\text{Frame}_t = (\text{uid}, e, t, \pi_t, \text{pub}_t, \text{blob}_A, \text{blob}_R, \sigma_t), \quad (4)$$

where uid , e , and t identify the pseudonymous admission chain, epoch, and round, π_t is the SHTPCMAIN proof, blob_A and blob_R are the encrypted Aggregator packages, and σ_t is the server-issued timing token and transport authentication material. The proof public statement contains

$$\text{pub}_t = (\text{policy}, e, t, c_{t-1}, c_a, \text{Com}_{\text{pri}}, \text{Com}_{\text{pay}}, \text{Com}_A, \text{Com}_R, \text{blob_hash}, \text{com}_{\text{sec}}). \quad (5)$$

The private witness contains the hidden coordinate, predecessor and anchor openings, the derived primary, commitment randomness, the clipped one-hot payload value, and the canonical share-content digests (d_A, d_R). The client computes

$$\text{blob_hash} = H(\text{blob}_A, \text{blob}_R, \text{ctx}_t)$$

and opens Com_A and Com_R to the same (primary, d_A, d_R, ctx_t) values used by C21. Each Aggregator decrypts only its own blob and accepts it only when the domain separator, context fields, channel label, length, deterministic expansion, and share-content digest match the proof-bound public binder. Thus the Shuffler checks proof and transport consistency, while Aggregators check channel-local package consistency. Neither role learns the hidden primary.

The identity commitment is

$$\text{com}_{\text{sec}} = \text{Poseidon}(\text{secret}, \text{uid}), \quad (6)$$

where com_{sec} is the public identity commitment, secret is the client secret sampled at enrollment, and uid is the registered user identifier. The secret remains client-side. The off-chain tag

$$\text{tag}_{\text{sec}} = \text{HMAC-SHA256}(\text{secret}, \text{uid}) \quad (7)$$

where tag_{sec} is an off-chain authentication tag. The tag is used only for enrollment and committee checks [84].

The Shuffler checks continuity by matching the next predecessor hash to the previously accepted current hash. The client therefore stores the salts needed to open the predecessor and the K -round anchor. For $K = 30$, the salt ring buffer costs $(K + 1) \cdot 32 \text{ B} = 992 \text{ B}$.

D. Distance Verification Circuit for Zero-Knowledge Proofs

The distance-verification circuit is the point where SHTPC's predicates become one accepted statement. It is designed to prevent a client from proving trajectory admissibility for one hidden primary while routing an aggregation payload for another, and to make all integer motion checks explicit inside the BN254 field.

The main circuit is parameterized by a compile-time window constant K . The evaluation instantiates two representative values, each with its own proving and verification parameters.

The Shuffler issues a signed timing token that binds $\Delta t = t_{\text{slot}} - t_{\text{last}}(\text{uid})$ to $(\text{uid}, \text{epoch}, w)$. Clients prove against the Shuffler-issued $\text{step_dt_sq} = \Delta t^2$ and are unable to choose or inflate their own Δt . The evaluated circuit is a fixed-slot artifact. Accepted non-gap reports use the configured $\Delta t_{\text{slot}} = 60 \text{ s}$, and session gaps re-enter warmup instead of enlarging the anchor window. A variable-slot deployment must replace the fixed-slot C14 derivation with a Shuffler-signed window-level cap. The defaults are $\Delta t_{\text{min}} = 1 \text{ s}$ and $G = 30 \text{ min}$.

The public statement contains predecessor, current, and anchor location hashes, the squared step duration, tier and mode speed caps, anchor caps, Com_{pri} , Com_{pay} , Com_A , Com_R , blob_hash , secret commitment, modeset commitment, and the public mode tag. The witness contains the hidden coordinates, salts, primary-cell opening, share-content digest openings used by the payload-binding relation, secret material, and hidden mode selector bits.

The circuit has six constraint groups. C1–C3 bind salted location hashes. C4–C5 enforce movement constraints for each step. C6–C7 bind payload and identity. C8–C10 enforce mode membership and tag consistency. C11–C16 enforce ADWC anchor caps and cap/tier consistency. C17–C20 derive the hidden primary cell from the current coordinate. C21 binds that primary to Com_{pri} , Com_{pay} , Com_A , and Com_R under the same report context.

Full formulas are given in Section IV-H. All integer comparisons use $\text{LessEqThan}(60)$ gadgets with explicit Num2Bits range checks on coordinates, squared distances, and public caps. Section IV-H1 gives the exact semantics. The implementation uses arity-3 Poseidon checks, and the evaluated relation compiles to 3,547 measured R1CS constraints with 16 public inputs and 25 private inputs.

Constraint C21 is the key relation that separates SHTPC from composing a trajectory proof with a VDAF-style check.

C17–C20 derive $\text{primary} = W \cdot \lfloor y/s \rfloor + \lfloor x/s \rfloor$ from the hidden coordinate. C21 then commits that same primary in Com_{pri} , Com_{pay} , Com_A , and Com_R under report context ctx , with d_A, d_R as canonical A/R share-content digests.

Because C21 shares a circuit with C4–C5 and C11–C16, the accepted trajectory proof is bound to share material for the same primary cell. An out-of-circuit composition cannot enforce this linkage if the two proofs use independently chosen hidden primaries. The full binding formulas appear in Section IV-H, and the ablation in Section VI-B shows that removing C21 drops A6 rejection to zero.

The circuit proves openings for the predecessor, current, and anchor hashes. The Shuffler state machine enforces

$$\text{hash}_{\text{prev}}^{(w+1)} = \text{hash}_{\text{curr}}^{(w)} \quad (8)$$

and strictly increasing round indices through c_w . The state machine therefore prevents re-salting an old coordinate from replaying an accepted chain state.

E. Enrollment and Warmup

SHTPC cannot certify that the first committed location is ground truth, so enrollment and warmup define where the cryptographic claim starts. The design uses the same circuit path during warmup to avoid a weaker bootstrap mode, while withholding payloads until the chain has accumulated enough plausible state.

SHTPC uses one circuit path for bootstrap and steady-state proofs. For round w , the anchor index is

$$\text{anchor_idx}(w) = \max(0, w - K), \quad (9)$$

where $\text{anchor_idx}(w)$ is the index of the anchor state, w is the current round, and K is the compile-time window length. When $w < K$, the anchor is the genesis commitment c_0 . Anchor constraints are never skipped.

Warmup is fixed to $N_{\text{warm}} = \max(K, 2)$ for enrollment and re-enrollment. During warmup, submissions are verified and advance chain state, but are withheld from aggregation. A session break occurs when the Shuffler-issued timing interval exceeds threshold G , set to 30 min by default. After a break, the client re-enters warmup. Mode-only and tier-changing re-enrollment rules are specified in Section IV-G.

F. Differential Privacy Layer

After Shuffler acceptance, the Decoder applies SVT/Laplace to the clipped count vector with sensitivity $\Delta f = 1$, as defined in §III-A. In the default $T = 10$ configuration, $\varepsilon_{\text{total}} = 1.0$ and $\varepsilon_{\text{round}} = 0.1$, split into $\varepsilon_{\text{SVT}} = 0.03$ and $\varepsilon_{\text{val}} = 0.07$.

For each round, the Decoder scans fixed grid cells in deterministic order, releases cells whose noisy counts clear the public threshold, and stops at the grid end or the public release cap. The released object is the threshold-filtered noisy count vector and released cell set, not the raw aggregate. Unspent validation budget remains unused, and protocol-control metadata is excluded from the ε -DP accounting.

G. Protocol Operations

The steady-state report path keeps admission, aggregation, and release separate. After Shuffler acceptance, a report is split into additive shares and forwarded to Aggregators A and R , then the Decoder reconstructs the aggregate and applies SVT/Laplace before releasing the DP heatmap.

Re-enrollment deliberately returns to warmup instead of inheriting full admission rights. A mode-only change rotates `mode_tag` and keeps `modeset_commitment`, while a tier change may update `tier_vmax_sq` and recompute the `modeset_commitment`.

Timing is server-issued to avoid client-selected motion budgets. The timing token binds $(uid, epoch, w, t_{last}, t_{slot}, \Delta t^2, \nu)$, and submission verification checks the signature, freshness, uid/epoch/round binding, nonce, and equality with the proof public input. Requests with $\Delta t < \Delta t_{min}$ are rejected, while gaps above G force warmup re-entry. The client stores (ℓ_w, r_w) for accepted states so the next proof can open the predecessor and the $\max(0, w + 1 - K)$ anchor. Reset starts a new epoch and never reuses salts.

H. Circuit Constraint Details

The circuit enforces the following constraints. C1–C20 cover trajectory, state, identity, mode, ADWC, and hidden-cell encoding. C21 is the in-circuit payload-binding relation. Table IV summarizes the constraint groups, and §IV-H1 states the integer semantics used by the comparisons.

TABLE IV
COMPRESSED CIRCUIT CONSTRAINT SUMMARY.

Group	Purpose
C1–C3	Salted Poseidon commitments for predecessor, current, and anchor locations.
C4–C5, C11–C16	Per-step speed bounds, ADWC anchor caps, cap/tier consistency, and selected-mode bounds.
C6–C10	Hidden-primary commitment, identity binding, modeset membership, and mode-tag consistency.
C17–C20	Coordinate-to-primary derivation with quotient and remainder constraints for cell size s , width W , and height H .
C21	Same-primary binding from Com_{pri} to Com_{pay} , Com_A , and Com_R under report context <code>ctx</code> .

The arity-3 Poseidon implementation with explicit integer checks compiles to 3,547 RICS constraints in the evaluated circuit.

1) *Integer Semantics and Range Constraints*: The BN254 scalar field has no native integer ordering, so every compared value is range-checked before use. Coordinates are signed 24-bit integers, coordinate differences are signed 25-bit integers, squared distances and public caps are 50-bit integers, `step_dt_sq` is 24-bit, and `tier_vmax_sq` is 16-bit. These bounds keep all compared values below 2^{60} . The circuit uses `Num2Bits` for coordinates, differences, distances, caps, and C17–C20 quotient/remainder witnesses, then implements all inequalities with `LessEqThan(60)`. C17–C20 enforce $x_2 = q_x s + r_x$, $y_2 = q_y s + r_y$, and $primary = q_y W + q_x$ with $0 \leq r_x, r_y < s$, $0 \leq q_x < W$, and $0 \leq q_y < H$. The evaluated grid uses $s = 100m$, $W = 100$, and $H = 100$.

These range checks prevent field wraparound in the distance and cap comparisons and make primary-cell derivation unique without exposing primary to the Shuffler.

2) *Private-Payload-Binding Tests*: The implementation additionally mutates each public binder, ciphertext blob, channel label, replayed package, and report-context field after a valid baseline submission. All negative cases are rejected deterministically by C17–C21, Shuffler `blob_hash` recomputation, or Aggregator-side format and context validation.

V. SECURITY AND THEORETICAL ANALYSIS

The implementation evidence used for this analysis is the evaluated Groth16 relation with 3,547 constraints and a 16/25 public/private input split.

Theorem 1 covers A1, A3, and A5. Lemma 6 covers A2a. Theorem 2 covers A6. A4 is bounded by the deployment proposition, while A2b and false genesis remain deployment-scope limitations.

Theorem 1 (Game 1 Envelope Admission Soundness). *For any PPT adversary $\mathcal{A}$ making at most $q = \text{poly}(\lambda)$ submission queries, if a submission is accepted by the SHTPC verifier and Shuffler state machine, and assuming Groth16 knowledge soundness, collision resistance of the hash relations inside the circuit, ciphertext-package authenticity, SRV-TS, and Shuffler State Consistency (SCS), then except with $\text{negl}(\lambda)$ probability the extracted witness satisfies the following properties.*

E1 Continuity constraints for each step (C4/C5). Under SRV-TS,

$$\|p_i - p_{i-1}\|^2 \leq \min(\text{tier_vmax_sq}, \text{mode_vmax_sq}) \cdot \text{step_dt_sq}.$$

E2 Chain continuity. *The predecessor hash matches the previous accepted Shuffler state.*

E3 Anchor Distance Window Continuity (C11–C16).

$$\|p_i - p_{\max(0, i-K)}\|^2 \leq \min(\text{tier_anchor_cap_sq}, \text{cap_policy_sq})$$

and

$$\text{cap_policy_sq} \leq \text{tier_anchor_cap_sq}.$$

For q submissions,

$$\begin{aligned} \text{Adv}_{\mathcal{A}}^{\text{traj}} &\leq q \cdot \text{Adv}^{\text{KS-Groth16}} \\ &\quad + c_p q \cdot \text{Adv}^{\text{CR-Poseidon}} \\ &\quad + q \cdot \text{Adv}^{\text{CR-MiMC7}} \\ &\quad + \text{negl}(\lambda). \end{aligned} \tag{10}$$

Here c_p is the constant number of Poseidon commitment openings checked by SHTPCMAIN per accepted submission.

Proof. Let Win be the event that $\mathcal{A}$ outputs an accepted submission whose extracted witness violates at least one of E1–E3. The proof uses a sequence of games. In G_0 , the challenger runs the real SHTPC acceptance experiment, so

$$\Pr[\text{Win}_0] = \text{Adv}_{\mathcal{A}}^{\text{traj}}.$$

In G_1 , the challenger aborts if the Groth16 knowledge-soundness extractor fails on any of the at most q accepting

SHTPCMAIN proofs. By knowledge soundness and a union bound,

$$|\Pr[\text{Win}_1] - \Pr[\text{Win}_0]| \leq q \cdot \text{Adv}^{\text{KS-Groth16}}.$$

In G_1 , every accepted proof has an extracted witness satisfying the circuit relation. In G_2 , the challenger aborts if any two distinct Poseidon openings collide across the location commitments and the commitment relations used by the continuity and payload-binding checks. Since SHTPCMAIN checks a constant number c_P of Poseidon commitment openings per accepted submission,

$$|\Pr[\text{Win}_2] - \Pr[\text{Win}_1]| \leq c_P q \cdot \text{Adv}^{\text{CR-Poseidon}}.$$

In G_3 , the challenger aborts if two distinct Shuffler chain-state preimages collide under MiMC7. Collision resistance gives

$$|\Pr[\text{Win}_3] - \Pr[\text{Win}_2]| \leq q \cdot \text{Adv}^{\text{CR-MiMC7}}.$$

In G_3 , each accepted submission has a unique extracted witness satisfying the circuit relation and a unique chain-state preimage. Therefore C4/C5 and SRV-TS imply E1, the Shuffler state machine under SCS implies E2, and the ADWC constraints C11–C16 imply E3. Thus $\Pr[\text{Win}_3] = 0$. Summing the three hop bounds yields Eq. (10). $\square$

Lemma 5 (Same-Primary Binding). *Let π be an accepted SHTPCMAIN proof with public binders $(\text{Com}_{\text{pri}}, \text{Com}_{\text{pay}}, \text{Com}_A, \text{Com}_R)$ and report context ctx . Under Groth16 knowledge soundness and Poseidon collision resistance, the extracted witness contains a unique hidden coordinate (x, y) such that $\text{primary} = W[y/s] + [x/s]$ is the same value committed in all four binders, except with negligible probability.*

Proof. C17–C20 constrain the witness primary to equal $W[y/s] + [x/s]$ derived from the hidden coordinate (x, y) . C6 binds this primary to Com_{pri} . C21 binds the same Com_{pri} to Com_{pay} , Com_A , and Com_R under ctx . Under Groth16 knowledge soundness, extraction succeeds except with $q \cdot \text{Adv}^{\text{KS-Groth16}}$. Under Poseidon collision resistance, no adversary opens any binder to a different primary with non-negligible probability. Therefore all four binders identify the same hidden primary cell except with negligible probability. $\square$

Theorem 2 (Game 2 End-to-End Payload Consistency). *Assume Groth16 knowledge soundness for $\mathcal{R}_{\text{SHTPC}}$, binding or collision resistance for the commitment and digest functions, ciphertext-package authenticity, deterministic share expansion, Shuffler blob_hash recomputation, and Aggregator-side format and context validation. For any PPT adversary that outputs an accepted and forwarded report frame Frame_t , except with negligible probability the decrypted Aggregator share packages decode to a clipped one-hot contribution whose primary cell equals the primary cell derived inside $\mathcal{R}_{\text{SHTPC}}$ from the hidden coordinate witness. The theorem proves internal proof-to-payload consistency only. It does not prove that the coordinate is a true physical location.*

Proof. First, acceptance of π_t implies, by Groth16 knowledge soundness, an extracted witness containing

$(x_t, y_t, \text{primary}_t, \dots)$ that satisfies $\mathcal{R}_{\text{SHTPC}}$, except with negligible probability. Constraints C17–C20 force $\text{primary}_t = W[y_t/s] + [x_t/s]$. Lemma 5 then gives the same primary_t across Com_{pri} , Com_{pay} , Com_A , and Com_R .

Second, the public binders and blob_hash bind the extracted primary to the report payload and the two channel-local share-content digests. Opening the same binder to an alternative primary $p'_t \neq \text{primary}_t$ or replacing a proof-bound ciphertext package requires either a commitment or digest collision or a failure of ciphertext-package authenticity.

Third, each Aggregator accepts only if its decrypted package matches the proof-bound context, channel label, deterministic expansion rule, and share-content digest. Therefore the only share packages that can reach the Decoder are the packages bound to primary_t in the accepted proof statement. A successful proof-to-payload substitution would need to violate at least one of the assumptions above. Hence the decoded clipped one-hot contribution is internally consistent with the hidden coordinate-derived primary, except with negligible probability. $\square$

Lemma 6 (Secret Binding). *Under the assumptions of Theorem 1, any accepted submission opens the registered per-user secret commitment checked by C7, except with negligible probability.*

Proof. C7 enforces $\text{Poseidon}(\text{secret}, \text{uid}_{\text{field}}) = \text{secret_commitment}$ as a circuit constraint. Under Groth16 knowledge soundness, every accepted proof extracts a witness satisfying C7. Under the collision resistance of Poseidon, the extracted $(\text{secret}, \text{uid}_{\text{field}})$ pair is unique to the registered secret_commitment. The EA registry binds each secret_commitment to exactly one enrollment identity. The Shuffler state machine enforces chain continuity per *uid*. Therefore an adversary who does not know secret cannot produce an accepting proof under a different *uid*'s commitment. The advantage is bounded by $q \cdot \text{Adv}^{\text{KS-Groth16}} + q \cdot \text{Adv}^{\text{CR-Poseidon}} + \text{negl}(\lambda)$. $\square$

Payload-binding boundary. A successful A6 substitution across the gap between proof and payload requires one of four failures. The adversary must break the extracted payload-binding relation inside the circuit, produce a commitment collision, bypass Shuffler blob_hash checking, or make Aggregator-side validation accept malformed share material.

Tier and Sybil soundness are composed from circuit binding to the public tier/modeset inputs, registry matching against threshold-valid EA attestations, Ed25519 Existential Unforgeability under Chosen-Message Attack (EUF-CMA), CA3, and per-anchor credential cost.

Proposition 1 (Tier-Attestation and Sybil-Cost Deployment Bound). *Any accepted tier must match a threshold-valid EA attestation. If each Sybil identity at tier τ requires t non-colluding anchors with expected per-anchor cost C_{anchor} and credential cost $c_{\text{VC}}(\tau)$, then under budget B ,*

$$m \leq \left\lfloor \frac{B}{t \cdot C_{\text{anchor}} + c_{\text{VC}}(\tau)} \right\rfloor. \quad (11)$$

This is a deployment-cost upper bound under the stated attestation and budgeting assumptions, not a cryptographic soundness theorem.

Theorem 3 (Game 3 Coordinate-and-Primary Hiding). *Consider two executions that have identical public parameters, role corruptions, uid/epoch topology, timing bins, message-size classes, accept/reject traces, and all other Shuffler-visible leakage L_S , but may differ in hidden coordinates and primary cells. Against an honest-but-curious Shuffler or any one corrupted Aggregator, the distinguishing advantage for the hidden coordinates and primary cells is bounded by*

$$\begin{aligned} \Delta_{\text{priv}} \leq & n \cdot \text{Adv}^{\text{ZK-Groth16}} \\ & + c_H n \cdot \text{Adv}^{\text{HIDE-Poseidon}} \\ & + n \cdot \text{Adv}^{\text{IND-CPA/AEAD}} \\ & + n \cdot \text{Adv}^{\text{IndexHide-Share}} \\ & + \text{negl}(\lambda), \end{aligned} \quad (12)$$

where n is the number of reports in the compared executions and c_H is the constant number of salted Poseidon commitments whose messages are hidden per report.

Proof. The proof is a standard hybrid over the corrupted server view. First, simulate SHTPCMAIN Groth16 proofs from the public statement. This simulation covers the entire circuit witness, including the trajectory witness, hidden-cell witness, and payload-binding witness. Second, replace the three location hashes and the public payload binders by salted Poseidon commitments to random messages. Third, replace Shuffler-routed ciphertexts with AEAD encryptions of dummy fixed-layout blobs. Finally, simulate any single corrupted Aggregator plaintext-share view by single-Aggregator index hiding. EA receives no trajectory data. The four hybrid hops give Eq. (12). $\square$

The SHTPC-violation mapping in the opening paragraph identifies which formal statement or deployment assumption covers each SHTPC-violation family.

Game 3 protects coordinates and the primary cell under the fixed-leakage scope in §III-C. A corrupted Shuffler observing per-uid accept/reject sequences can detect coarse mobility patterns, including sleep schedules, mode changes, and commute regularity. This leakage is discussed in §VII. It does not reveal coordinate values or primary cells. Each result bounds one component of the adversary’s advantage. Theorem 1 bounds acceptance of SHTPC trajectory-envelope violations. Theorem 2 bounds mismatches between proof and payload. Theorem 3 bounds coordinate-and-primary distinguishability under equal L_S . Game 4 covers the released heatmap, while enrollment is handled by the deployment proposition. A joint adversary’s advantage is therefore bounded by the sum of these game advantages plus the leakage excluded from $\varepsilon_{\text{total}}$.

VI. ADMISSION-BOUNDARY ENFORCEMENT EVALUATION

The evaluation follows the narrowed paper claims. It does not measure general location truthfulness or poisoning robustness. RQ1 tests whether SHTPC rejects proof-to-payload inconsistent reports. RQ2 tests whether stateful predecessor

and K -window checks enforce the public envelope. RQ3 quantifies what remains inferable from Shuffler-visible metadata L_S . RQ4 characterizes residual in-envelope and deployment-scope risks. RQ5 measures heatmap utility after admission and DP release. RQ6 measures proof-carrying execution cost and scalability. All rejection results should be read as boundary-enforcement results.

A. Experimental Setup

Datasets and design-space comparisons. The original-input utility evaluation uses T-Drive taxi trajectories (10,357 taxis and about 15M GPS points), GeoLife mixed-mode traces (182 users and more than 17K trajectories), and taxi trajectories from Porto and Rome. Each prepared real input contains $N=1000$ client records with bootstrap disabled and ten windows per record. The fifth input, Synthetic, contains $N=1000$ clustered random-walk records with the same ten-window format and generation seed 20260709. Boundary and session experiments also use T-Drive rush-hour and GeoLife session-gap subsets to stress dense vehicle mobility and re-enrollment behavior [1, 2, 85–89]. We compare SHTPC-FULL with NO-ADMISSION-CHECK, NO-ADWC, NO-PAYLOAD-BIND, STATEFUL-CONTEXTCOMMIT, and deployment-layer variants. STATEFUL-CONTEXTCOMMIT tracks uid/epoch/round, predecessor state, report context, blob hashes, and share validation, but still separates trajectory admission from payload validation. Plaintext trusted filtering is reported only as an upper bound because it sees coordinates and primary cells. Table III explains why these are the relevant comparisons. Complete VDAF, TEE, or proof-of-location systems are not drop-in baselines for the same privacy and trust boundary. We therefore use them as analytical design points and measure the closest low-cost composition that keeps coordinate hiding and stateful context binding.

For RQ5, the shared heatmap harness also includes mechanism-level Nebula-style and EIFFeL-style adapters. The Nebula-style adapter models report sampling, shuffling, thresholding, and debiasing. The EIFFeL-style adapter models the single-round range and norm validation outcome. These adapters do not reproduce native deployments of the complete upstream systems. Utility comparisons are therefore limited to the evaluated adapters under the shared fixed protocol.

Boundary-violation suites and configuration. The controlled harness covers A1 teleportation, A2a secret substitution, A3 gradual drift, A5 replay or chain-state reuse, and A6 proof-payload substitution. A4 Sybil creation is analyzed by Proposition 1. A2b transfer and false genesis are deployment-scope risks. The adaptive suite maximizes accepted pollution under public parameters and covers reset/warmup abuse, false genesis followed by plausible motion, near-boundary drift, and proof-payload mismatch. Unless stated otherwise, the headline configuration uses the following settings.

- Controlled boundary-violation headlines use $N=50$ clients and $T=10$ rounds. The fixed utility comparison uses $N=1000$ and $T=10$.
- $\varepsilon_{\text{total}}=1.0$, $\Delta t=60$ s, and $K \in \{6, 30\}$.
- mode table {walk, bike, vehicle, transit} with speeds {3, 10, 33, 40} m/s.

- $\text{cap}_{\text{policy}}=3.3 \text{ km}$ ($K=6$) and 16.5 km ($K=30$), satisfying $\text{cap}_{\text{policy_sq}} \leq \text{tier_anchor_cap_sq}$.
- 100 m cells over a 10 000 m domain.

The count payload is L_1 -clipped to one primary cell. Headline rejection uses $K=6$. The $K=30$ setting is a long-window stress profile.

Metrics and implementation. TPR is the malicious rejection rate. Pipeline FRR is circuit rejection plus session withholding plus DP-SVT exclusion on benign traces. RMSE/Pearson measure count distortion, Jaccard measures released-set stability, false hotspot probability and HS@k measure ranking pollution. The prototype uses Python microservices and Circom/snarkjs Groth16 inside Docker. RQ6 reports end-to-end scale from the validated five-dataset, three-seed aggregate. Randomized sweeps use three seeds unless stated otherwise.

Hardware and software. All experiments ran on an Apple M4 MacBook Pro with 10 CPU cores and 16 GB of host memory. The scale evaluation used two concurrent jobs. Docker was allocated 10 CPUs and 8.22 GB of memory. The software environment used Python 3.10.20, Node.js 20.20.2, Circom 2.1.9, snarkjs 0.7.6, and Docker Server 29.1.3 on Linux arm64 containers.

Original-input utility. The RQ5 evaluator reads the five prepared inputs above directly, uses seeds 101, 202, and 303, and applies one fixed release configuration to SHTPC and the evaluated baselines. These original-input runs supply all utility values in Table VI and Figure 5.

Safe-trajectory protocol smoke. The clean Docker protocol suite uses the evaluated circuit with protocol-isolation trajectories. It replaces timestamp gaps and out-of-envelope jumps with safe in-envelope trajectories. This layer isolates proof generation, state transitions, reconstruction, and DP release across the five dataset-labelled runs. It is not the source of the RQ5 utility values.

Full Groth16 scale. The scale matrix exercises five datasets and seeds 101, 202, and 303 at $N=1000$, with six warmup and ten evaluation rounds under full snarkjs Groth16 proving. It uses the same protocol-isolation trajectory and therefore measures end-to-end protocol execution rather than original-input utility. The validated aggregate binds all 15 dataset-seed records to one circuit and execution configuration. It retains every functionally passing unit in protocol totals and admits a unit to timing statistics only when its maximum inter-round gap is at most 7,200 seconds. Each dataset must retain at least two timing-eligible units.

B. RQ1 Proof-to-Payload Consistency

RQ1 asks whether SHTPC rejects reports whose proof and forwarded payload are inconsistent. The main target is A6, where a valid trajectory proof is paired with payload material for a different primary cell. We also report the other SHTPC violation families in the same matrix to show which mechanism guards each admission boundary. Table V reports the within-pipeline ablation on GeoLife under the default controlled violation configuration. Each row removes one mechanism and checks which report family becomes admissible.

TABLE V
WITHIN-PIPELINE ADMISSION ABLATION. ENTRIES ARE MALICIOUS REJECTION RATES FOR SHTPC-VIOLATION FAMILIES UNDER THE GEOLIFE DEFAULT CONTROLLED CONFIGURATION.

Scheme	A1	A2a	A3	A5	A6
SHTPC-FULL	1.00	1.00	1.00	1.00	1.00
No-ADWC	1.00	1.00	0.00	1.00	1.00
No-PAYLOAD-BIND	1.00	1.00	1.00	1.00	0.00
STATEFUL-CONTEXTCOMMIT	1.00	1.00	1.00	1.00	0.00
No-ADMISSION-CHECK	0.00	0.00	0.00	0.00	0.00

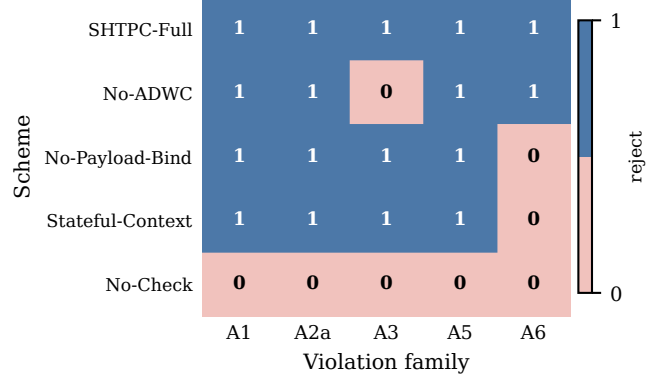


Fig. 2. Mechanism-level rejection matrix for Table V. A value of 1 means that the violation family is rejected by the admission layer, and 0 means that it is admitted. The matrix shows that ADWC is needed for A3 and same-primary binding is needed for A6.

SHTPC-FULL rejects all modeled A1, A2a, A3, A5, and A6 violations, so Table V and Figure 2 should be read as admission-consistency results. Primary-cell hiding is a formal Game 3 claim under equal L_S , and post-admission utility is evaluated separately in RQ5. No-ADMISSION-CHECK accepts every modeled violation. No-ADWC loses A3 because gradual drift is the anchor-window invariant. No-PAYLOAD-BIND and STATEFUL-CONTEXTCOMMIT both admit A6 because trajectory and payload statements are separated.

A6 is also tested through a private payload-binding harness that mutates hidden primary commitments, report payload binders, A/R share-content commitments, ciphertext packages, channel labels, replay state, and report context. Negative cases are rejected across proof, transport, and aggregation validation.

C. RQ2 Stateful Envelope Boundary

RQ2 asks how statefulness affects predecessor, replay, and K -window consistency. The measured STATEFUL-CONTEXTCOMMIT baseline represents a strong context-binding composition without in-relation primary equality. It rejects stale predecessor replay, gradual drift, timing-token replay, and context substitution, but it still admits A6 because the trajectory proof can be valid for one hidden primary while the payload is routed to another. SHTPC-FULL is the only coordinate-hiding row in Table V that rejects A6 because the trajectory-derived primary and payload primary are constrained inside one relation.

Boundary sharpness. Boundary sweeps separate the two continuity predicates. For E1, vehicle-tier proof generation

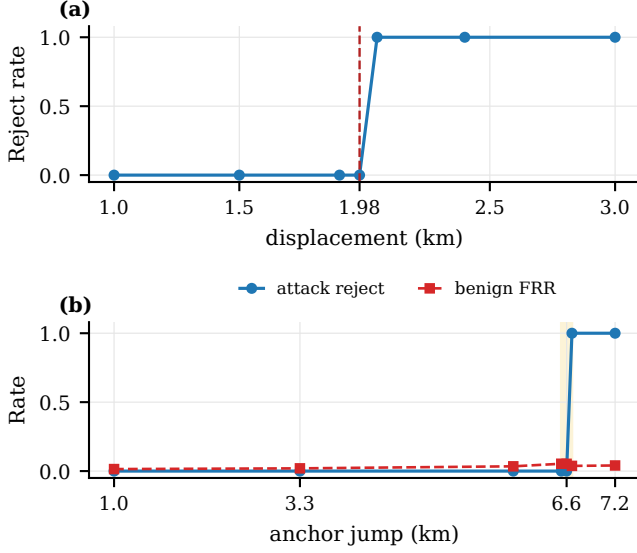


Fig. 3. Boundary sweeps for the two trajectory predicates. Panel (a) plots per-step rejection around the vehicle-tier displacement cap. Panel (b) plots attack rejection and benign FRR around the ADWC anchor boundary.

accepts displacements up to $\tau_{\text{vehicle}}\Delta t = 1980$ m and rejects larger displacements. For E3, the T-Drive ADWC sweep at $K=6$, vehicle tier, and $\text{cap}_{\text{policy}} = 3.3$ km transitions near 6.6 km in the plotted attack-jump parameterization because the sweep moves the adversarial endpoint across both sides of the anchor boundary. The underlying admission predicate remains the 3.3 km anchor-distance cap. The observed transition reflects the windowed anchor-distance boundary rather than a single-step speed cap.

A3 is evaluated as a *parameterized ADWC admission check*. Two deployment profiles are used, $K=6$ ($\text{cap}_{\text{policy}}=3.3$ km) and $K=30$ ($\text{cap}_{\text{policy}}=16.5$ km). For $K=6$, TPR= 1.00 across all five drift biases (0.20–1.00). For $K=30$, TPR ranges from 0.80 (bias= 0.20) to 0.94 (bias= 0.40), reflecting the larger window’s softer resolution for each step. $K=30$ is a stress test rather than the recommended deployment. Tightening $\text{cap}_{\text{policy}}$ from $1.67\times$ to $0.67\times$ default raises $K=30$ TPR from 0.89 to 0.99 in the policy-cap sensitivity sweep.

(τ, τ_2) sensitivity and session-gap baseline. A (τ, τ_2) sweep over $\{1, \dots, 5\}^2$ (75 runs, 3 seeds per pair) confirms DP-parameter invariance. Mean TPR 0.981 (range 0.951–1.000) and session-withholding rate exactly 0.192 across all pairs (circuit-FRR 0). This is expected because proof verification precedes the DP/SVT release stage.

D. RQ3 Shuffler-Visible Metadata Leakage

RQ3 asks what behavior remains inferable from L_S . The adversary observes only Shuffler-visible metadata such as chain length, timing, topology, and accept/reject histories. On synthetic calibrated traces with 500 users, 30 rounds, delay mean 1.5 rounds, and three seeds, timing-only and chain-topology views reach AUC 0.748 ± 0.005 and 0.751 ± 0.004 , nearly matching full metadata (0.751 ± 0.003). Accept/reject history is weaker but still above random (0.679 ± 0.011). On

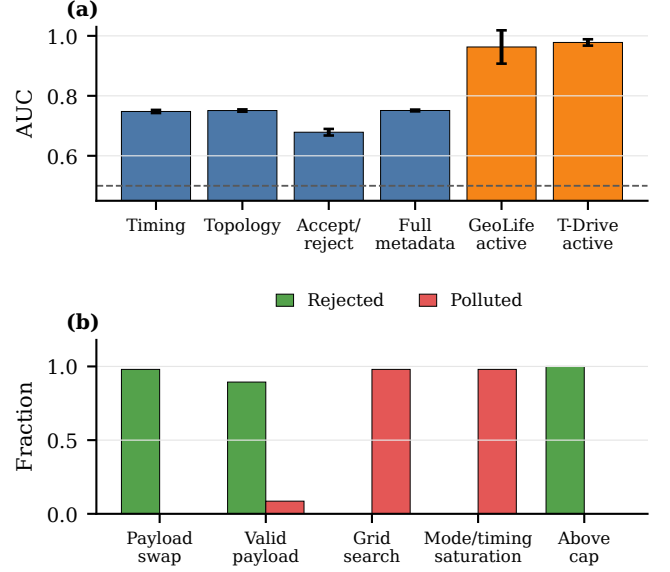


Fig. 4. Dense evaluation summary. Panel (a) reports metadata-only inference from Shuffler-visible L_S , with the dashed line marking random AUC. Panel (b) reports adaptive boundary outcomes and separates rejected attempts from accepted polluted outcomes.

50-user, 30-round real fragments, active-period inference is highly predictable on GeoLife (0.963 ± 0.056) and T-Drive (0.978 ± 0.011), while lightweight session linking remains near random. Thus SHTPC hides coordinates and primary cells under Game 3, but not participation, chain growth, timing regularity, or accept/reject traces.

E. RQ4 Residual In-Envelope and Deployment-Scope Risks

RQ4 asks what remains after the cryptographic envelope is enforced. SHTPC bounds a malicious client’s spatial influence per round to at most $\tau_u\Delta t$ and per K -window to $\text{cap}_{\text{policy}}$. Under the default vehicle profile, where $\tau_u\Delta t = 1980$ m, one accepted report can target at most 1,229 cells, or about 12.3% of the grid. Under $K=6$ and $\text{cap}_{\text{policy}} = 3.3$ km, the per-identity per-window reach is at most 3,409 cells, or 34.09% of the grid. This reach bound limits where accepted malicious mass can appear. It does not multiply down the malicious clients’ total L_1 mass, because an adaptive adversary may concentrate its clipped contribution on one reachable cell.

Proposition 2 (Accepted Malicious Mass and Support Bounds). *Let ρ be the malicious-client fraction in a reporting window. Let C_{clip} be the per-report L_1 clipping constant. Let $R_u \subseteq \mathcal{G}$ be the set of cells reachable by malicious client u under the public $\text{cap}_{\text{policy}}$, with $|R_u| \leq A_{\text{cap}}$. For the normalized heatmap perturbation $\Delta\bar{h}$ caused by accepted malicious reports,*

$$\|\Delta\bar{h}\|_1 \leq \rho C_{\text{clip}}, \quad \|\Delta\bar{h}\|_\infty \leq \rho C_{\text{clip}},$$

and

$$|\text{supp}(\Delta\bar{h})| \leq \left| \bigcup_{u \in M} R_u \right| \leq \min(|\mathcal{G}|, |M|A_{\text{cap}}).$$

Thus A_{cap} bounds the spatial support of accepted malicious influence, but it does not reduce the worst-case L_1 mass.

Proof. Each accepted malicious report has total count weight at most C_{clip} , which gives the normalized L_1 bound ρC_{clip} . All malicious clients may target the same reachable cell, so the same value is the worst-case per-cell bound. The support bound follows from taking the union of reachable sets R_u . $\square$

Joint adaptive adversary. SHTPC enforces an envelope, not ground truth. In the joint adaptive suite ($N=100$, 10% malicious clients, $T=50$, $K=6$, $\text{cap}_{\text{policy}}=3.3$ km), payload-decoupled attempts are rejected in 98% of malicious cell-rounds and admit no in-envelope contribution. Payload-bound in-envelope attempts are accepted by design (490/500 cell-rounds), while the same strategy at $1.01\times$ the ADWC cap is rejected in all 500 malicious cell-rounds. False genesis and A2b credential/device transfer remain deployment-scope risks. With $G=30$ min, inter-window gaps exceeding G are 1.10% on T-Drive and 2.12% on GeoLife.

F. RQ5 Utility After Admission and DP Release

RQ5 asks how much utility is lost after adding SHTPC before the unchanged SVT/Laplace DP release. The headline protocol fixes $\varepsilon=5.0$ and the post-admission release threshold $\tau=2.0$ for every dataset, uses $N=1000$, $T=10$, 10% A1 attackers, top-50 released-set Jaccard, and seeds 101, 202, and 303. It does not substitute per-dataset tuned settings. Among the evaluated fixed-configuration adapters, the Nebula-style adapter has the highest mean Jaccard on every dataset [11, 15].

Table VI and Figure 5 report mean Jaccard for SHTPC and the Nebula-style adapter. The respective values are 0.671/0.561 on T-Drive, 0.878/0.366 on GeoLife, 0.305/0.298 on Porto, 0.772/0.664 on Rome, and 0.135/0.128 on Synthetic. The absolute gains are therefore +0.110, +0.512, +0.007, +0.108, and +0.007, respectively. Porto and Synthetic provide only small gains, so the result supports an observed positive direction under the fixed protocol rather than a uniformly large margin. All 15 observed dataset-seed paired differences favor SHTPC, and the smallest is +0.00239. Each dataset has only three seeds, so we do not claim statistical significance for the small Porto and Synthetic differences. SHTPC also has zero honest false rejection and complete modeled malicious rejection on every dataset in this comparison. A separate GeoLife sensitivity sweep found admission TPR 1.00 at each tested ε , consistent with proof verification preceding the release stage. It is supporting evidence rather than the headline utility comparison. As shown in RQ4, accepted in-envelope mass can still create false hotspots, which remains a deployment risk.

G. RQ6 Cost and Scalability

RQ6 asks whether proof-carrying admission is practical at per-report granularity. The evaluated Groth16 relation has 3,547 constraints, 16 public inputs, and 25 private inputs. It passes all 22 static checks and clean protocol smoke on 5 datasets. The strict three-seed $N=1000$ aggregate verifies all 15 dataset-seed units, with 225,000 client proofs generated and

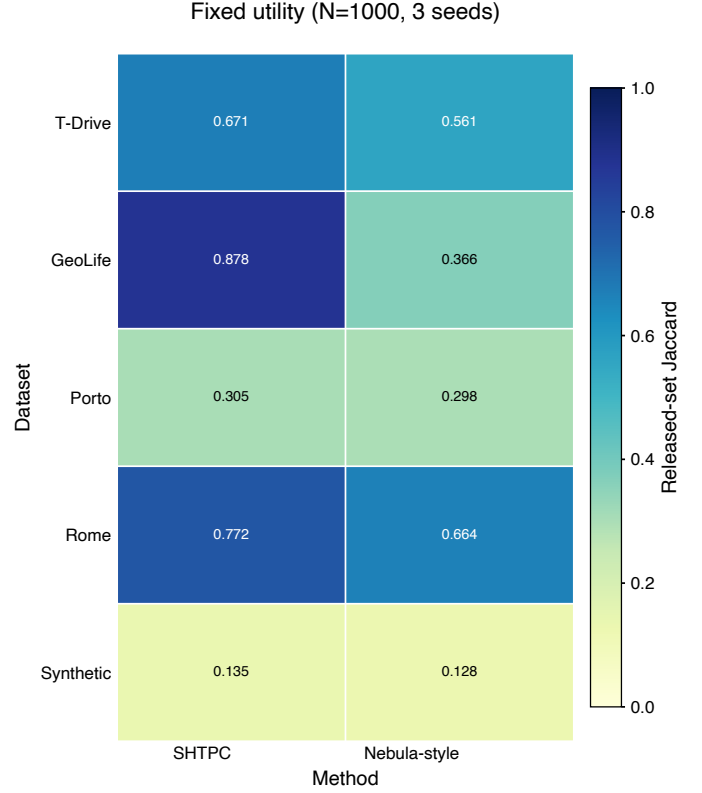


Fig. 5. Fixed-configuration released-set Jaccard at $N=1000$ and $T=10$, averaged over seeds 101, 202, and 303. Both methods use $\varepsilon=5.0$. SHTPC uses post-admission release threshold $\tau=2.0$. Table VI reports standard deviations, absolute deltas, FRR, and MRR.

Shuffler-verified and 0 proof failures. Across 150 evaluation rounds, routes A and R receive 150,000 and 150,000 reports, respectively. All 150 reconstructions and 150 DP releases pass.

TABLE VI
FIXED FAIR FIVE-DATASET UTILITY COMPARISON AT $N=1000$, THREE SEEDS, $\varepsilon=5.0$, AND $\tau=2.0$. NEBULA-STYLE DENOTES THE MECHANISM-LEVEL ADAPTER IN THE SHARED HEATMAP HARNESS. HIGHER JACCARD IS BETTER.

Dataset	SHTPC $\pm$ std	Nebula-style $\pm$ std	Δ	FRR	MRR
T-Drive	0.671 $\pm$ 0.026	0.561 $\pm$ 0.000	+0.110	0.000	1.000
GeoLife	0.878 $\pm$ 0.044	0.366 $\pm$ 0.002	+0.512	0.000	1.000
Porto	0.305 $\pm$ 0.006	0.298 $\pm$ 0.003	+0.007	0.000	1.000
Rome	0.772 $\pm$ 0.009	0.664 $\pm$ 0.003	+0.108	0.000	1.000
Synthetic	0.135 $\pm$ 0.006	0.128 $\pm$ 0.002	+0.007	0.000	1.000

TABLE VII
GROTH16 PROTOCOL EXECUTION AT $N=1000$ OVER THREE SEEDS. ALL FUNCTIONALLY VALID UNITS CONTRIBUTE TO PROOF TOTALS. TIMING EXCLUDES UNITS WITH AN INTER-ROUND GAP ABOVE 7,200 SECONDS; ENTRIES REPORT MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER THE REMAINING TIMING-ELIGIBLE UNITS.

Dataset	Verified	Timing n	Proofs/s	Wall time (h)
T-Drive	45,000	3	1.072 $\pm$ 0.127	3.921 $\pm$ 0.439
GeoLife	45,000	3	0.994 $\pm$ 0.027	4.195 $\pm$ 0.113
Porto	45,000	3	0.967 $\pm$ 0.020	4.308 $\pm$ 0.091
Rome	45,000	2	0.967 $\pm$ 0.031	4.310 $\pm$ 0.138
Synthetic	45,000	2	0.983 $\pm$ 0.009	4.240 $\pm$ 0.039

Table VII separates functional completion from timing eligibility. Rome seed 101 and Synthetic seed 101 each contain an inter-round gap above 62,962 seconds caused by host suspension. Their functional records remain in every proof, route, reconstruction, and release total, but the two units are excluded only from timing statistics under the 7,200-second rule. The remaining 13 units, with 2 exclusions, average 1.000 ± 0.069 proofs/s and 4.183 ± 0.248 hours per unit (mean $\pm$ sample standard deviation across timing-eligible units). Thus the prototype establishes reproducible full-matrix execution with no proof or release failure, but it does not yet provide interactive per-report latency or optimized production throughput.

VII. DISCUSSION AND LIMITATIONS

The TCB consists of the Groth16 setup, EA Ed25519 anchor keys, AEAD/channel authentication, and Poseidon/MiMC7 collision resistance over BN254 [90]. Aggregator collusion degrades input privacy to the public DP release while admission consistency still holds. A compromised Groth16 setup mainly threatens soundness.

Metadata is an intentional boundary rather than an implementation artifact. The Shuffler sees L_S , including chain topology, timing bins, mode/tier metadata, message-size classes, and accept/reject outcomes. These signals can support operational admission control, but they are not protected by the public heatmap’s $\varepsilon_{\text{total}}$ -DP accounting and can reveal behavioral regularity. Thus SHTPC should not be read as trajectory anonymity or report unlinkability.

Several mitigations can reduce L_S in deployments, including fixed-rate reporting, batching, dummy warmup traffic, coarser timing bins, reset cooldowns, and externally auditable Shuffler logs. These mitigations trade latency, bandwidth, and operational complexity for less metadata exposure. They are not part of the core SHTPC claim unless they are explicitly

included in the leakage function and evaluated under the same Game 3 comparison.

SHTPC enforces an envelope, not ground truth. Fully in-envelope false reports remain admissible when they satisfy predecessor continuity, ADWC, timing tokens, and same-primary payload binding. False genesis and credential/device transfer (A2b) remain outside the circuit guarantee. These risks can preserve all post-enrollment invariants and therefore require operational controls such as proof-of-location at enrollment, device attestation, secure-element sealing, behavioral scoring, quotas, reset cooldowns, and enrollment throttling.

SHTPC is best suited to managed heatmap deployments where enrollment, device credentials, and Shuffler accountability are controllable. It is a poor fit for fully anonymous trajectory systems, deployments that need report unlinkability, or settings that require security against a completely untrusted Shuffler without replicated logs or external audit.

A. Artifact and Reproducibility Details

The proof-to-payload claim depends on implementation-level binding, so the artifact index records the protocol interface and source metadata rather than only aggregate metrics. It identifies the Circom source, R1CS and verification key hashes, public-signal order, C21 formulas, coordinate and grid mapping, Poseidon and MiMC7 parameters, frame and AEAD context construction, share-content digests, seed-compressed share expansion, and Aggregator checks. It also separates the static checks, five-dataset protocol smoke, fixed utility, publication figure, execution configuration, 15 unit records, and hash-bound aggregate. Scale results enter the manuscript only through the validated aggregate that checks every unit record, functional counter, timing rule, and circuit/configuration binding.

The four real trajectory datasets are obtained directly from their official providers under the terms recorded in `DATA_NOTICE.md`. Neither raw GPS records nor row-level prepared real trajectories are redistributed. The artifact contains preparation code, a non-row-level source manifest, random seeds, and aggregate records, tables, and figures. The Synthetic input is generated by repository code with a fixed seed.

The checked-in Groth16 proving and verification material is treated as a development reproducibility artifact. Any change to the public-signal order, K , grid parameters, hash domains, or payload-binder formula requires regenerating or revalidating it. This material is not evidence of a production multi-party ceremony. A production deployment would require an

approved setup process, hashes, and ceremony metadata. A compromised setup mainly threatens admission soundness.

B. Ethics, Governance, and Misuse

SHTPC hides raw coordinates and primary cells from the allowed single-server views, but it deliberately keeps persistent pseudonymous admission state. This state can reveal participation frequency, activity periods, mode or tier changes, resets, and accept or reject patterns. These metadata can support operational accountability, but they can also enable behavioral profiling if retained or queried without governance.

The evaluation is a secondary analysis of provider-released T-Drive, GeoLife, Porto, and Rome trajectories plus repository-generated Synthetic records. It does not recruit participants or collect new location traces. Reproducers must obtain the real datasets from their official sources and comply with their license, citation, non-de-anonymization, and redistribution terms. The artifact does not release raw or row-level prepared real trajectories.

Deployments should therefore define consent, retention, revocation, access control, and audit policies before using SHTPC. Shuffler logs and enrollment records should be minimized, time-limited, and auditable. Operators should document that the system does not provide trajectory anonymity or physical presence proof. High-risk deployments, including law-enforcement-adjacent uses, should require stronger external oversight because the admission state can become a surveillance choke point even when raw coordinates remain hidden.

VIII. CONCLUSION

This paper addresses hidden trajectory-payload mismatch in private heatmap aggregation through the SHTPC admission property. Its key idea is to bind predecessor continuity, K -window reachability, hidden-primary derivation, and payload equality inside one zero-knowledge relation before aggregation separates the report across roles. The evaluated Groth16 relation contains 3,547 constraints, 16 public inputs, and 25 private inputs. The evaluation shows that SHTPC rejects the modeled envelope and proof-payload violations admitted by its ablations and by STATEFUL-CONTEXTCOMMIT. The fixed comparison also yields positive observed Jaccard differences on all five datasets, although two differences are small. Across 15 $N=1000$ units, the full-path matrix verifies 225,000 proofs with no proof, reconstruction, or DP-release failure. Its roughly 4.183-hour mean unit runtime supports reproducible execution rather than interactive latency. These results support managed heatmap systems with stateful admission and explicit governance. They do not extend to physical-presence proof, metadata anonymity, fully in-envelope false reports, or a malicious Shuffler without external accountability.

Future Work. Future work should evaluate stronger enrollment, device attestation, auditable Shufflers, larger adaptive sweeps, and universal-setup proof-system ports such as PLONK [71].

REFERENCES

- [1] Y. Zheng et al., “GeoLife: A collaborative social networking service among user, location and trajectory,” *IEEE Data Engineering Bulletin*, vol. 33, no. 2, pp. 32–40, 2010.
- [2] J. Yuan et al., “T-drive: Driving directions based on taxi trajectories,” in *Proceedings of the 18th SIGSPATIAL International Conference on Advances in Geographic Information Systems (GIS)*, 2010, pp. 99–108.
- [3] C. Dwork et al., “Calibrating noise to sensitivity in private data analysis,” in *Proceedings of the Theory of Cryptography Conference (TCC)*, 2006, pp. 265–284.
- [4] A. Bittau et al., “Prochlo: Strong privacy for analytics in the crowd,” in *Proceedings of the 26th ACM Symposium on Operating Systems Principles (SOSP)*, 2017, pp. 441–459.
- [5] A. Cheu et al., “Distributed differential privacy via shuffling,” in *Proceedings of EUROCRYPT 2019*, 2019, pp. 375–403.
- [6] K. Bonawitz et al., “Practical secure aggregation for privacy-preserving machine learning,” in *Proceedings of the 24th ACM Conference on Computer and Communications Security (CCS)*, 2017, pp. 1175–1191.
- [7] H. Corrigan-Gibbs et al., “Prio: Private, robust, and scalable computation of aggregate statistics,” in *Proceedings of the 14th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2017, pp. 259–282.
- [8] R. Barnes et al., “Verifiable distributed aggregation functions,” Internet Engineering Task Force, Internet-Draft draft-irtf-cfrg-vdaf-20, Jun. 2026, Work in Progress.
- [9] B. Balle et al., “The privacy blanket of the shuffle model,” in *Proceedings of CRYPTO 2019*, 2019, pp. 638–667.
- [10] N. O. Tippenhauer et al., “On the requirements for successful GPS spoofing attacks,” in *Proceedings of the 18th ACM Conference on Computer and Communications Security (CCS)*, 2011, pp. 75–86.
- [11] A. S. Shamsabadi et al., “Nebula: Efficient, private and accurate histogram estimation,” in *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2025, pp. 498–512.
- [12] R. A. Popa et al., “VPriv: Protecting privacy in Location-Based vehicular services,” in *18th USENIX Security Symposium (USENIX Security 09)*, Montreal, Quebec: USENIX Association, Aug. 2009, pp. 335–350.
- [13] J. Balasch et al., “PrETP: Privacy-Preserving electronic toll pricing,” in *19th USENIX Security Symposium (USENIX Security 10)*, Washington, DC: USENIX Association, Aug. 2010, pp. 63–78.
- [14] C. Troncoso et al., “PriPAYD: Privacy-friendly pay-as-you-drive insurance,” *IEEE Transactions on Dependable and Secure Computing*, vol. 8, no. 5, pp. 742–755, 2011.
- [15] A. Roy Chowdhury et al., “EIFFeL: Ensuring integrity for federated learning,” in *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2022, pp. 2535–2549.
- [16] J. Groth, “On the size of pairing-based non-interactive arguments,” in *Proceedings of EUROCRYPT 2016*, 2016, pp. 305–326.
- [17] P. Blanchard et al., “Machine learning with adversaries: Byzantine tolerant gradient descent,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [18] D. Yin et al., “Byzantine-robust distributed learning: Towards optimal statistical rates,” in *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018, pp. 5650–5659.
- [19] X. Cao et al., “FLTrust: Byzantine-robust federated learning via trust bootstrapping,” in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2021.
- [20] T. D. Nguyen et al., “FLAME: Taming backdoors in federated learning,” in *Proceedings of the 31st USENIX Security Symposium*, 2022, pp. 1415–1432.
- [21] C. Dwork et al., *The Algorithmic Foundations of Differential Privacy*. Foundations and Trends in Theoretical Computer Science, 2014, vol. 9, pp. 211–407.
- [22] P. Kairouz et al., “Advances and open problems in federated learning,” *Foundations and Trends in Machine Learning*, vol. 14, no. 1–2, pp. 1–210, 2021.

- [23] I. Mironov, "Rényi differential privacy," in *Proceedings of the 30th IEEE Computer Security Foundations Symposium (CSF)*, 2017, pp. 263–275.
- [24] J. Dong et al., "Gaussian differential privacy," *Journal of the Royal Statistical Society: Series B*, vol. 84, no. 1, pp. 3–37, 2022.
- [25] S. L. Warner, "Randomized response: A survey technique for eliminating evasive answer bias," *Journal of the American Statistical Association*, vol. 60, no. 309, pp. 63–69, 1965.
- [26] Ú. Erlingsson et al., "RAPPOR: Randomized aggregatable privacy-preserving ordinal response," in *Proceedings of the 21st ACM Conference on Computer and Communications Security (CCS)*, 2014, pp. 1054–1067.
- [27] R. Bassily et al., "Local, private, efficient protocols for succinct histograms," in *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, 2015, pp. 127–135.
- [28] T. Wang et al., "Locally differentially private protocols for frequency estimation," in *Proceedings of the 26th USENIX Security Symposium*, 2017, pp. 729–745.
- [29] N. Wang et al., "Collecting and analyzing multidimensional data with local differential privacy," in *Proceedings of the 35th IEEE International Conference on Data Engineering (ICDE)*, 2019, pp. 638–649.
- [30] M. E. Andrés et al., "Geo-indistinguishability: Differential privacy for location-based systems," in *Proceedings of the 20th ACM Conference on Computer and Communications Security (CCS)*, 2013, pp. 901–914.
- [31] K. Chatzikokolakis et al., "Broadening the scope of differential privacy using metrics," in *Proceedings of the 13th Privacy Enhancing Technologies Symposium (PETS)*, 2013, pp. 82–102.
- [32] N. E. Bordenabe et al., "Optimal geo-indistinguishable mechanisms for location privacy," in *Proceedings of the 21st ACM Conference on Computer and Communications Security (CCS)*, 2014, pp. 251–262.
- [33] M. C. González et al., "Understanding individual human mobility patterns," *Nature*, vol. 453, pp. 779–782, 2008.
- [34] C. Song et al., "Limits of predictability in human mobility," *Science*, vol. 327, no. 5968, pp. 1018–1021, 2010.
- [35] Y.-A. de Montjoye et al., "Unique in the crowd: The privacy bounds of human mobility," *Scientific Reports*, vol. 3, p. 1376, 2013.
- [36] L. Sweeney, "*k*-anonymity: A model for protecting privacy," *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, vol. 10, no. 5, pp. 557–570, 2002.
- [37] M. Gruteser et al., "Anonymous usage of location-based services through spatial and temporal cloaking," in *Proceedings of the 1st International Conference on Mobile Systems, Applications and Services (MobiSys)*, 2003, pp. 31–42.
- [38] B. Hoh et al., "Preserving privacy in GPS traces via uncertainty-aware path cloaking," in *Proceedings of the 14th ACM Conference on Computer and Communications Security (CCS)*, 2007, pp. 161–171.
- [39] M. Terrovitis et al., "Privacy preservation in the publication of trajectories," in *Proceedings of the 9th IEEE International Conference on Mobile Data Management (MDM)*, 2008, pp. 65–72.
- [40] M. E. Nergiz et al., "Towards trajectory anonymization: A generalization-based approach," *Transactions on Data Privacy*, vol. 2, no. 1, pp. 47–75, 2009.
- [41] V. Primault et al., "The long road to computational location privacy: A survey," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 3, pp. 2772–2793, 2019.
- [42] H. Wang et al., "PrivTrace: Differentially private trajectory synthesis by adaptive Markov models," in *Proceedings of the 32nd USENIX Security Symposium*, 2023, pp. 1649–1666.
- [43] Y. Du et al., "LDPTTrace: Locally differentially private trajectory synthesis," *Proceedings of the VLDB Endowment*, vol. 16, no. 8, pp. 1897–1909, 2023.
- [44] Ú. Erlingsson et al., "Amplification by shuffling: From local to central differential privacy via anonymity," in *Proceedings of the 30th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2019, pp. 2468–2479.
- [45] V. Feldman et al., "Hiding among the clones: A simple and nearly optimal analysis of privacy amplification by shuffling," in *Proceedings of the 62nd IEEE Annual Symposium on Foundations of Computer Science (FOCS)*, 2021, pp. 954–964.
- [46] B. Balle et al., "Private summation in the multi-message shuffle model," in *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2020, pp. 657–676.
- [47] B. Ghazi et al., "On the power of multiple anonymous messages: Frequency estimation and selection in the shuffle model of differential privacy," in *Proceedings of the 40th Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT)*, 2021, pp. 463–488.
- [48] J. Bell et al., "Secure single-server aggregation with (Poly)Logarithmic overhead," in *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2020, pp. 1253–1269.
- [49] D. Boneh et al., "Lightweight techniques for private heavy hitters," in *Proceedings of the 2021 IEEE Symposium on Security and Privacy (S&P)*, 2021, pp. 762–776.
- [50] J. Bell et al., "ACORN: Input validation for secure aggregation," in *Proceedings of the 32nd USENIX Security Symposium*, 2023, pp. 4805–4822.
- [51] M. Rathee et al., "ELSA: Secure aggregation for federated learning with malicious actors," in *Proceedings of the 44th IEEE Symposium on Security and Privacy (S&P)*, 2023, pp. 1961–1979.
- [52] S. Xu et al., "Camel: Communication-efficient and maliciously secure federated learning in the shuffle model of differential privacy," in *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2024, pp. 243–257.
- [53] M. Fang et al., "Local model poisoning attacks to Byzantine-robust federated learning," in *Proceedings of the 29th USENIX Security Symposium*, 2020, pp. 1605–1622.
- [54] E. Bagdasaryan et al., "How to backdoor federated learning," in *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020, pp. 2938–2948.
- [55] L. Melis et al., "Exploiting unintended feature leakage in collaborative learning," in *Proceedings of the 40th IEEE Symposium on Security and Privacy (S&P)*, 2019, pp. 691–706.
- [56] M. L. Psiaki et al., "GNSS spoofing and detection," *Proceedings of the IEEE*, vol. 104, no. 6, pp. 1258–1270, 2016.
- [57] M. Götzelmann et al., "Galileo open service navigation message authentication: Preparation phase and drivers for future service provision," *NAVIGATION: Journal of the Institute of Navigation*, vol. 70, no. 3, navi.572, 2023.
- [58] R. Shokri et al., "Quantifying location privacy," in *Proceedings of the 2011 IEEE Symposium on Security and Privacy (S&P)*, 2011, pp. 247–262.
- [59] R. Shokri et al., "Protecting location privacy: Optimal strategy against localization attacks," in *Proceedings of the 2012 ACM Conference on Computer and Communications Security (CCS)*, 2012, pp. 617–627.
- [60] M. Rosenberg et al., "Zk-creds: Flexible anonymous credentials from zkSNARKs and existing identity infrastructure," in *Proceedings of the 44th IEEE Symposium on Security and Privacy (S&P)*, 2023, pp. 790–808.
- [61] D. Rathee et al., "ZEBRA: SNARK-based anonymous credentials for practical, private and accountable on-chain access control," *Cryptology ePrint Archive*, Report 2022/1286, 2022.
- [62] J. Allen et al., "An algorithmic framework for differentially private data analysis on trusted processors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.
- [63] J. Day et al., "SPEcTRE: Spot-checked private ecash tolling at roadside," in *Proceedings of the 10th ACM Workshop on Privacy in the Electronic Society (WPES)*, 2011, pp. 61–68.
- [64] S. Meiklejohn et al., "The phantom tollbooth: Privacy-preserving electronic toll collection in the presence of driver collusion," in *20th USENIX Security Symposium (USENIX Security 11)*, San Francisco, CA: USENIX Association, Aug. 2011.
- [65] W. de Jonge et al., "Privacy-friendly electronic traffic pricing via commits," in *Formal Aspects in Security and Trust (FAST 2008)*, ser. Lecture Notes in Computer Science, vol. 5491, 2009, pp. 143–161.
- [66] F. D. Garcia et al., "Cell-based privacy-friendly roadpricing," *Computers & Mathematics with Applications*, vol. 65, no. 5, pp. 774–785, 2013.

- [67] S. Goldwasser et al., “The knowledge complexity of interactive proof systems,” *SIAM Journal on Computing*, vol. 18, no. 1, pp. 186–208, 1989.
- [68] E. Ben-Sasson et al., “Succinct non-interactive zero knowledge for a von Neumann architecture,” in *Proceedings of the 23rd USENIX Security Symposium*, 2014, pp. 781–796.
- [69] B. Parno et al., “Pinocchio: Nearly practical verifiable computation,” in *Proceedings of the 34th IEEE Symposium on Security and Privacy (S&P)*, 2013, pp. 238–252.
- [70] E. Ben-Sasson et al., “Zerocash: Decentralized anonymous payments from bitcoin,” in *Proceedings of the 2014 IEEE Symposium on Security and Privacy (S&P)*, 2014, pp. 459–474.
- [71] A. Gabizon et al., *PLONK: Permutations over Lagrange-bases for oecumenical noninteractive arguments of knowledge*, Cryptology ePrint Archive, Report 2019/953, 2019.
- [72] E. Ben-Sasson et al., “Aurora: Transparent succinct arguments for RICS,” in *Proceedings of EUROCRYPT 2019*, 2019, pp. 103–128.
- [73] S. Setty, “Spartan: Efficient and general-purpose zkSNARKs without trusted setup,” in *Proceedings of CRYPTO 2020*, 2020, pp. 704–737.
- [74] A. Kothapalli et al., “Nova: Recursive zero-knowledge arguments from folding schemes,” in *Proceedings of CRYPTO 2022*, 2022, pp. 359–388.
- [75] B. Bünz et al., “Bulletproofs: Short proofs for confidential transactions and more,” in *Proceedings of the 39th IEEE Symposium on Security and Privacy (S&P)*, 2018, pp. 315–334.
- [76] R. S. Wahby et al., “Doubly-efficient zkSNARKs without trusted setup,” in *Proceedings of the 39th IEEE Symposium on Security and Privacy (S&P)*, 2018, pp. 926–943.
- [77] A. Kate et al., “Constant-size commitments to polynomials and their applications,” in *Proceedings of ASIACRYPT 2010*, 2010, pp. 177–194.
- [78] L. Grassi et al., “Poseidon: A new hash function for zero-knowledge proof systems,” in *Proceedings of the 30th USENIX Security Symposium*, 2021, pp. 519–535.
- [79] M. R. Albrecht et al., “MiMC: Efficient encryption and cryptographic hashing with minimal multiplicative complexity,” in *Proceedings of ASIACRYPT 2016*, 2016, pp. 191–219.
- [80] L. Grassi et al., “Reinforced concrete: A fast hash function for verifiable computation,” in *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2022, pp. 1323–1335.
- [81] H. Wen et al., “Practical security analysis of zero-knowledge proof circuits,” in *Proceedings of the 33rd USENIX Security Symposium*, 2024, pp. 1471–1487.
- [82] A. Shamir, “How to share a secret,” *Communications of the ACM*, vol. 22, no. 11, pp. 612–613, 1979.
- [83] J. R. Douceur, “The sybil attack,” in *Proceedings of the 1st International Workshop on Peer-to-Peer Systems (IPTPS)*, 2002, pp. 251–260.
- [84] M. Bellare et al., “Keying hash functions for message authentication,” in *Proceedings of CRYPTO 1996*, 1996, pp. 1–15.
- [85] J. Yuan et al., “Driving with knowledge from the physical world,” in *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2011, pp. 316–324.
- [86] Y. Zheng et al., “Understanding mobility based on GPS data,” in *Proceedings of the 10th International Conference on Ubiquitous Computing (UbiComp)*, 2008, pp. 312–321.
- [87] Y. Zheng et al., “Mining interesting locations and travel sequences from GPS trajectories,” in *Proceedings of the 18th International Conference on World Wide Web (WWW)*, 2009, pp. 791–800.
- [88] L. Moreira-Matias et al., *Taxi service trajectory – prediction challenge, ECML PKDD 2015*, UCI Machine Learning Repository, DOI: 10.24432/C55W25, 2015.
- [89] L. Bracciale et al., *CRAWDAD dataset roma/taxi (v. 2014-07-17)*, CRAWDAD, DOI: 10.15783/C7QC7M, 2014.
- [90] V. Shoup, “Lower bounds for discrete logarithms and related problems,” in *Proceedings of EUROCRYPT 1997*, 1997, pp. 256–266.